When Explanations Outlive Their Data: Faithfulness Decoupling in Graph-Attention MARL Fleet Dispatch under Telemetry Degradation

A Master's Dissertation

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MSc in Artificial Intelligence

By

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DECLARATION

I declare that this dissertation is my own original work carried out under the supervision of Farhan S. Ujager and that all sources and references have been acknowledged appropriately. This dissertation has not been submitted previously for academic credit at De Montfort University, Dubai, or any other institution.

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Name: Hongwei Lin

Date: ______________________________

ACKNOWLEDGMENT

I would like to express my sincere gratitude to my supervisor, Dr Farhan S. Ujager, for his valuable support and guidance throughout this research. His advice, encouragement and careful feedback helped me refine the experiment, interpret the findings responsibly and complete this dissertation.

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LIST OF ABBREVIATIONS

Abbreviation	Full form
AoI	Age of Information
CI	Confidence Interval
CTDE	Centralized Training with Decentralized Execution
DEF	Decision-level Explanation Faithfulness
GAT	Graph Attention Network
GNN	Graph Neural Network
GxI	Gradient x Input
LOO	Leave-One-Out
MAPPO	Multi-Agent Proximal Policy Optimization
MARL	Multi-Agent Reinforcement Learning
MLP	Multilayer Perceptron
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
SUMO	Simulation of Urban Mobility
WAMSN	Weighted Attention Mass on Stale Nodes
XAI	Explainable Artificial Intelligence
XRL	Explainable Reinforcement Learning

ABSTRACT

Graph-attention multi-agent reinforcement learning can expose attention weights as candidate explanations of fleet-dispatch decisions. However, an attention map may remain visible after its supporting vehicle telemetry becomes stale. This dissertation asks whether raw graph-attention weights can be trusted as explanations under clean and degraded telemetry.

The experiment simulates 20 taxis in SUMO. Tunnel entry triggers an observation-layer outage that freezes an affected taxi's last valid position and speed for 10, 20, 30, or 60 seconds while SUMO retains the current traffic state. This produces paired clean and degraded observations for the same decision. GAT policies trained on clean data or with 30-second outages are evaluated across three independent training seeds. Decision-level explanation faithfulness (DEF) compares attention-ranked nodes with type-matched, action-protected random controls. WAMSN separately measures attention assigned to stale vehicle information. Further checks cover leave-one-out rankings, attention extraction, action type, and random telemetry-loss triggers.

The results do not validate raw attention as a reliable explanation. Clean corrected DEF remains near zero and varies across checkpoints. Longer outages increase stale-data exposure, but paired attention and DEF shifts reverse direction for one training seed. The main associations are dominated by no-op decisions and do not reproduce for dispatch actions. The seed-dependent pattern also appears under random triggering and in both training regimes. Therefore, the tested attention weights do not provide consistent evidence of freshness or decision relevance.

The dissertation proposes a freshness-aware explanation audit framework that reports freshness and action composition separately, applies action-aware faithfulness tests, checks the attention extraction rule, and requires consistent evidence across checkpoints. Attention remains an internal diagnostic when these checks fail. The framework governs explanation release; it does not repair the model. Its demonstrative application returns WITHHOLD for both model families and all six checkpoints.

Keywords: explainable reinforcement learning; graph attention; multi-agent reinforcement learning; fleet dispatch; telemetry degradation; faithfulness.

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Chapter 1: Introduction

1.1 Context

Fleet dispatch is relational. A taxi's useful action depends on nearby vehicles, open passenger requests, road conditions, and competition for the same demand. Graph neural networks represent these relationships directly, and graph attention networks (GATs) add learned weights over neighboring nodes (Scarselli et al., 2009; Velickovic et al., 2018). Because the weights are easy to display, they are often read as an explanation of which vehicles or requests influenced a decision.

That reading is convenient, but it is not automatically valid. Prior work has shown that an attention weight can be weakly related to the evidence that actually changes a model's output (Jain and Wallace, 2019; Serrano and Smith, 2019; Wiegreffe and Pinter, 2019; Liu et al., 2022). The issue becomes more serious in an operational system because the input itself may no longer describe the current world.

1.2 Problem statement

Vehicle telemetry can be interrupted by tunnels and other signal-loss areas. During an interruption, a dispatch platform may keep the last received position and speed. The reading remains available, but its Age of Information (AoI) grows (Kaul, Yates and Gruteser, 2012). An attention map can therefore look complete even though some nodes represent old information. The interface reveals where the model placed attention, but it does not reveal whether that information is fresh or whether removing it would change the decision.

This creates an assurance problem. An operator may read a complete attention map as a trustworthy reason even when its evidence is stale, and stable dispatch performance cannot show whether the explanation is valid. This dissertation focuses on explanation reliability rather than increasing the number of pickups. It tests whether raw graph-attention weights provide reliable evidence about a decision under clean and degraded telemetry.

This dissertation uses **faithfulness decoupling** to describe the risk that an explanation remains available and visually plausible after its supporting data have become stale, without a reliable link between the displayed weights and the evidence that changes the decision. One possible pattern is declining faithfulness while dispatch performance remains stable. The core question is whether attention provides reliable evidence under the tested conditions.

1.3 Research questions

The central research question is:

Can raw graph-attention weights be trusted as explanations of fleet-dispatch decisions when vehicle telemetry becomes stale?

Four operational questions provide the evidence needed to answer it:

1. **RQ1:** Is graph attention a faithful explanation under clean telemetry?
2. **RQ2:** When tunnel-triggered outages occur, does attention move toward stale vehicle nodes?
3. **RQ3:** As outage duration increases, does the relationship between attention, stale-data exposure, and decision relevance remain reliable?

4. **RQ4:** Does training the policy under telemetry degradation make the explanation response more reliable?

The construct-validity audit supports RQ1-RQ3. It is not a separate primary problem. Its purpose is to check that the faithfulness metric measures hidden information rather than the accidental deletion of an available action.

1.4 Objectives

The measurable objectives are to:

- build a reproducible SUMO fleet-dispatch benchmark with training, validation, and held-out test demand;
- train clean and degradation-aware policies across three independent seeds;
- trigger signal loss from tunnel entry while applying the resulting freeze at the observation boundary;
- compare clean telemetry with fixed observation-layer outages of 10, 20, 30, and 60 seconds;
- test whether the paired result remains similar when telemetry loss is triggered randomly rather than by one fixed tunnel location;
- measure explanation faithfulness with action-aware, type-matched counterfactual controls;
- check whether the faithfulness evaluator is sensitive to an LOO perturbation ranking and quantify the resolution lost through top-k overlap;
- measure how much attention is assigned to stale vehicle information;
- test whether conclusions change across attention layers, heads, rollout, and the query row used as the explanation;
- treat the trained policy, rather than each individual decision, as the unit for judging whether a finding is consistent; and
- turn the collected evidence into an explicit ELIGIBLE, WITHHOLD, or INCOMPLETE explanation-release decision.

1.5 Main findings

Overall, **this study does not validate raw attention weights as reliable explanations**.

Under clean telemetry, type-matched DEF remains close to its random control and varies across the six checkpoints. A leave-one-out perturbation ranking gives a larger positive DEF in every checkpoint, showing that the evaluator can respond to a decision-relevant ranking even though raw attention does not provide a stable result.

Longer outages increase AoI-weighted stale exposure for every checkpoint. The paired response is less consistent: checkpoints trained with seeds 42 and 43 assign more attention to stale nodes on average, while those trained with seed 44 assign less. Paired DEF also changes direction across checkpoints. A random-trigger check shows a similar overall pattern, making a tunnel-only explanation less plausible without establishing a universal degradation effect.

The action audit narrows the conclusion. Among scorable decisions with at least one available request, 93.0% select no-op. The main duration and exposure associations do not reproduce within the smaller dispatch stratum. Attention results also change with the layer, head, aggregation rule, and query row used to construct the explanation.

Degradation-aware training does not remove this variation. The sampled policy distributions outperform the declared lower bounds, but all six graph-attention checkpoints choose no-op in the deterministic diagnostic. The study therefore audits explanations for sampled stochastic decisions; it does not establish a deployable argmax dispatcher. Overall, freshness and faithfulness must be checked separately for every policy checkpoint before release.

1.6 Contributions

The dissertation makes four main contributions:

1. a paired clean/degraded benchmark that separates SUMO ground truth from the observation received by the policy and distinguishes the tunnel trigger from the observation-layer outage;
2. an action-aware, type-matched faithfulness protocol for graphs in which request nodes also define available actions, supported by LOO, overlap, and action-stratified diagnostics;
3. evidence across independently trained checkpoints, attention-extraction choices, and tunnel and random triggers showing that stale-data exposure is consistent but attention reallocation and faithfulness effects are not; and
4. a reproducible freshness-aware explanation audit framework that converts the evidence into explicit ELIGIBLE, WITHHOLD, and INCOMPLETE release decisions.

1.7 Dissertation structure

Chapter 2 reviews MARL dispatch, graph attention, explanation faithfulness, and AoI. Chapter 3 describes the simulator, models, degradation layer, metrics, and statistical design. Chapter 4 reports the results. Chapter 5 discusses what can and cannot be concluded. Chapter 6 closes with the practical implications for trustworthy fleet-dispatch explanations.

Chapter 2: Literature Review

This chapter reviews the literature needed to define the research gap. It first considers reinforcement learning for relational fleet dispatch, then examines explanations for sequential and graph-based decisions. It also reviews the debate over attention and the limits of faithfulness tests. The final sections connect these topics to stale telemetry and position the experiment.

2.1 Reinforcement learning for fleet dispatch

Fleet dispatch links vehicles, passenger requests, and a road network over time. A decision for one taxi changes the demand available to others, so a collection of independent single-agent policies can create competition or duplicated assignments. Lin et al. (2018) model city-scale fleet management as a cooperative multi-agent reinforcement-learning (MARL) problem. Qin, Zhu and Ye (2022) show that reinforcement learning is used for matching, repositioning, pricing, and route decisions, often under changing supply and demand.

MARL offers several ways to represent coordination. MADDPG learns decentralized actors with centralized critics (Lowe et al., 2017). QMIX factorizes a team value into agent values while preserving a monotonic relation to the joint action (Rashid et al., 2020). MAPPO uses the more familiar PPO objective with centralized training and decentralized execution; its empirical stability makes it a practical baseline for cooperative tasks (Yu et al., 2022). This dissertation adopts MAPPO because the research focus is the explanation attached to each taxi's local decision, not a new MARL algorithm.

Graphs provide a natural representation of the local dispatch problem. Nodes can represent the acting taxi, nearby taxis, and open requests; edges permit their information to be combined (Scarselli et al., 2009). A graph attention network (GAT) assigns learned weights while aggregating neighboring nodes (Velickovic et al., 2018). Actor-Attention-Critic similarly uses attention for communication in MARL (Iqbal and Sha, 2019). This makes GAT-MARL a suitable test case, not because it is assumed to be the best dispatch model, but because it matches the relational, multi-agent structure of fleet dispatch while exposing an attention channel that can be audited. These weights are easy to visualize, which makes them tempting to present as reasons for an action. Ease of display, however, is not evidence that the weights identify the information that caused the decision.

Recent dispatch methods continue to use relational structure. BMG-Q represents ride-pooling dispatch through a local bipartite matching graph (Hu, Feng and Li, 2025); CoopRide studies cooperation across city grids (Wang et al., 2025); and DualG-MARL combines vehicle-state and task graphs (Sha et al., 2026). These studies evaluate dispatch or scheduling quality rather than whether graph weights remain valid explanations when vehicle telemetry becomes stale. They provide useful task benchmarks, but not direct faithfulness benchmarks.

2.2 Explainability in reinforcement learning

Explainable reinforcement learning (XRL) covers more than a visual saliency map. An explanation may describe which state features mattered, summarize a policy, contrast the selected action with an alternative, or show the sequence of events that led to an outcome. Reviews by Heuillet, Couthouis and Diaz-Rodriguez (2021), Puiutta and Veith (2020), and

Milani et al. (2024) all stress that the method must match the user's question. A model developer debugging a policy and an operator checking one live action do not need the same output. The systematic taxonomy by Bekkemoen (2024) reaches the same conclusion across a larger body of recent XRL studies.

This dissertation concerns a local, post-decision operator question: *which visible vehicles or requests supported the selected action, whether dispatch or no-op?* Earlier XRL work offers several ways to answer this question. Greydanus et al. (2018) perturb image regions to visualize what changes an Atari policy. Madumal et al. (2020) use a causal model to generate contrastive explanations. Mott et al. (2019) introduce an attention bottleneck intended to expose what an RL agent uses. These approaches differ in mechanism, but they share an important lesson: a useful picture and a faithful account of model dependence are separate properties.

For an operational dashboard, readability and faithfulness must be considered separately. An explanation may be easy to read but weakly connected to the policy computation. A faithful explanation should change when information that matters to the action is removed or retained. This experiment tests that link through a controlled decision-level faithfulness test; it does not measure user preference for the attention map.

2.3 Is attention an explanation?

The modern debate began with Jain and Wallace (2019), who found that attention weights could be weakly correlated with other importance measures and that very different attention distributions could produce similar predictions. Wiegreffe and Pinter (2019) argued that this does not justify rejecting all attention explanations and proposed more careful tests. Serrano and Smith (2019) likewise showed that removing high-attention items can affect outputs, but that the relationship varies by model and task. The literature therefore supports neither a universal acceptance nor a universal rejection of attention. Each explanation claim needs an explicit definition and an appropriate test.

Attention became widely visible through Transformer models (Vaswani et al., 2017), but visibility also encouraged a broader interpretability claim than the mechanism alone can support. Lipton (2018) warns that the word *interpretability* often combines several different goals that should be evaluated separately.

Jacovi and Goldberg (2020) separate *plausibility*, which concerns whether an explanation looks reasonable to a person, from *faithfulness*, which concerns whether it reflects the model's actual reasoning process. This dissertation follows that distinction. It treats the attention weights as a candidate explanation and measures their decision relevance rather than accepting them simply because they are part of the actor.

Layer and head aggregation add another difficulty. A two-layer, multi-head network does not produce one unique attention map. Averaging heads, selecting a layer, taking a maximum, or composing attention across layers can produce different rankings. Abnar and Zuidema (2020) propose attention rollout and flow to account for information mixing across layers, and report stronger correlations with ablation and gradient importance than raw attention in Transformers. Although their model is not a GAT-MARL dispatcher, the methodological point transfers: an attention result is incomplete unless the aggregation rule is defined and its sensitivity is checked.

2.4 Explaining graph neural networks

GNN explanations may identify important features, nodes, edges, or subgraphs. GNNExplainer learns a compact subgraph and feature mask that preserves a prediction (Ying et al., 2019). PGExplainer parameterizes explanation generation so that it can be shared across examples (Luo et al., 2020). SubgraphX searches for important subgraphs using Shapley-value approximations (Yuan et al., 2021). These methods show that graph explanation is a distinct problem: removing one node can alter both its own features and the messages available to other nodes.

The survey by Yuan et al. (2023) organizes GNN explainers by target, mechanism, and scope. GraphFramEx goes further by comparing explanation methods under different user needs and combining necessity and sufficiency views (Amara et al., 2022). Its results show that no single method dominates every evaluation dimension. This supports the use of several checks in the present study: DEF tests counterfactual relevance, taxi-only rank correlation compares attention with leave-one-out effects, and stale-attention measures describe freshness exposure.

Most GNN explainability benchmarks study node or graph classification. Fleet dispatch differs because some graph nodes also define actions. A passenger request is both information and a selectable action, while a nearby taxi is context only. Deleting them creates different interventions. This action-linked graph structure motivates the construct-validity audit in Section 3.7.

2.5 Faithfulness evaluation and its pitfalls

Comprehensiveness and sufficiency are common perturbation measures. In the ERASER benchmark, comprehensiveness asks how much a prediction weakens when the explanation is removed, while sufficiency asks how well the selected evidence alone preserves it (DeYoung et al., 2020). Liu et al. (2022) similarly test attention by measuring faithfulness violations. These measures are useful because they connect an explanation to model behavior instead of visual appeal.

Model-agnostic methods such as LIME also use local perturbations to estimate which inputs support a prediction (Ribeiro, Singh and Guestrin, 2016). Their usefulness depends on whether the perturbed samples remain meaningful for the model and task.

Perturbation is not automatically valid. Removing input features may create samples unlike the data seen during training. ROAR addresses this issue by removing features and retraining the model, showing why a simple deletion test can confound attribution quality with distribution shift (Hooker et al., 2019). Adebayo et al. (2018) provide a different sanity check: an explanation should respond when model parameters or training labels are randomized. Alvarez-Melis and Jaakkola (2018) show that explanation methods can also be locally unstable. Together, these studies show that the evaluator itself must be audited.

The present action-deletion problem is related but more specific. If a selected request node is masked, its action logit becomes unavailable. A large margin change can then occur even if the request features carried no useful evidence. A random baseline that removes more request nodes than the attention top-k is not a fair comparison. Type matching, selected-action protection, and clamp logging are introduced to control this mechanism. A leave-one-out perturbation ranking acts as a positive control for comprehensiveness, while Gradient x Input provides a comparator that does not use attention coefficients.

The small graph creates a second limit. When k = 3 and only a few valid nodes exist, a random size-three set must overlap substantially with the explanation set. This compresses the possible difference between them. Reporting the exact expected overlap and a taxi-only rank correlation makes that limitation visible instead of treating every near-zero DEF as decisive evidence.

2.6 Age of Information and telemetry degradation

Age of Information (AoI) measures the time since the newest received update was generated (Kaul, Yates and Gruteser, 2012). It is widely used to study communication freshness and the effect of delayed state on estimation or control. The explanation question is different. A dispatch action and its attention map may remain available even when a vehicle reading has stopped updating. The operator can see a strong weight without knowing that its source is old.

This dissertation uses AoI only as a node-level freshness descriptor and as the weight inside WAMSN. The experiment manipulates the duration of an observation-layer outage triggered by tunnel entry, not AoI itself. It therefore makes no causal claim that a larger AoI value lowers faithfulness. The analysis instead asks whether stale exposure and attention-based explanation behavior provide consistent evidence under controlled outages.

2.7 Research gap and positioning

The literature supports four points: relational RL is appropriate for fleet dispatch; attention is easy to expose but disputed as an explanation; GNN faithfulness requires graph-aware perturbations; and stale state matters to operational control. Existing work has not brought these topics together as one explanation-assurance problem.

Recent graph-based dispatch studies provide the closest task comparison, but they answer a different question from this dissertation. Table 2.1 summarizes their relationship to the present study.

Table 2.1: Recent graph-based dispatch literature compared with this study

Study	Setting and method	Main evaluation	Difference from this study
BMG-Q (Hu, Feng and Li, 2025)	ride-pooling with a local bipartite matching graph	dispatch performance	does not audit graph weights under stale telemetry
CoopRide (Wang et al., 2025)	cooperative MARL across city grids	city-scale dispatch performance	does not test node-level explanation faithfulness
DualG-MARL (Sha et al., 2026)	state and task graphs for ride-sharing scheduling	scheduling performance	does not pair clean and degraded observations or audit freshness
This dissertatio n	local GAT-MAPPO taxi/request graph	explanation assurance	tests paired telemetry degradation, action-aware faithfulness, and release conditions

These methods are not treated as performance baselines because their tasks, action spaces, and data-generation procedures differ. They are benchmark literature for positioning the contribution: recent work improves relational dispatch, whereas this dissertation tests whether an exposed attention channel has enough evidence to be presented as an explanation.

Within the graph- and RL-explainability studies reviewed in Sections 2.2-2.5, no evaluation protocol was identified that combines all of the following:

- a graph-attention MARL action explained at decision level;
- paired clean and stale versions of the same operational observation;
- explicit measurement of attention placed on stale telemetry;
- a control for graph nodes that also remove available actions; and
- replication across independently trained checkpoints.

This thesis contributes an audit protocol and an empirical test rather than a new dispatch algorithm. SUMO supplies a controlled simulated state (Lopez et al., 2018). The GAT-MAPPO policy supplies a realistic attention channel. The research contribution is to test whether that channel gives a reproducible freshness-aware and decision-relevant explanation, and to state clearly when the evidence cannot support that claim.

Chapter 3: Methodology

3.1 Study design

The study uses a controlled simulation because the research question requires the same decision to be observed in two ways: once with current telemetry and once with degraded telemetry. A real operational log cannot provide this exact clean twin. SUMO provides the simulated physical state, while a separate observation layer controls what the policy receives.

The protocol is defined in version-controlled configuration files. It contains three stages: training, validation-based checkpoint selection, and held-out evaluation. Faithfulness sweeps are run only after the selected checkpoints are frozen.

3.2 SUMO environment and data

The road network is an OpenStreetMap extract of the Central Park area in Yubei, Chongqing. Roads marked as tunnels supply physically meaningful signal-loss triggers. SUMO simulates 20 taxis and 50 passenger requests during each 1,200-second episode. Taxi dispatch is controlled through TraCI, while SUMO continues to update the true position and speed of every vehicle.

Demand variants are separated into training, validation, and test splits. The policy is fitted on training demand. Candidate checkpoints are compared on validation demand. The final reported runs use held-out test demand. This separation prevents test results from influencing model selection.

SUMO is a simulator rather than a neutral source of observed data (Lopez et al., 2018). Its outputs depend on the chosen road network, generated demand, routes, vehicle behavior parameters, tunnel locations, and random seeds. The experiment is therefore not described as free from bias. Instead, it uses these assumptions consistently in a paired design. At each sampled decision, the clean and degraded observations share the same underlying SUMO state and trained-policy checkpoint; only the observation-layer telemetry differs. This controls the comparison well enough to isolate the effect of the implemented degradation within this scenario, but it does not make the simulated fleet representative of every real city.

3.3 Observation graph and action space

Each idle taxi receives a self-centered graph. Table 3.1 summarizes its node types, features, and roles.

Table 3.1: Observation graph node types and roles

Node type	Main features	Role
Self taxi	position, time, speed, AoI, position-valid flag	acting vehicle
Peer taxi	relative position, availability, distance, AoI	nearby fleet context
Passenger request	pickup/drop-off vectors, waiting time	candidate request and action

The graph contains up to five nearby taxis and five nearby requests. Features are normalized and request/taxi masks prevent padded nodes from receiving attention.

The discrete action space contains no-op plus one action for each visible passenger request. This one-to-one relationship is important: deleting a passenger-request node also removes the associated action, whereas deleting a peer-taxi node only hides information about that taxi.

Section 3.7 explains how the faithfulness audit controls this asymmetry. Figure 3.1 shows the local graph and the request-to-action mapping.

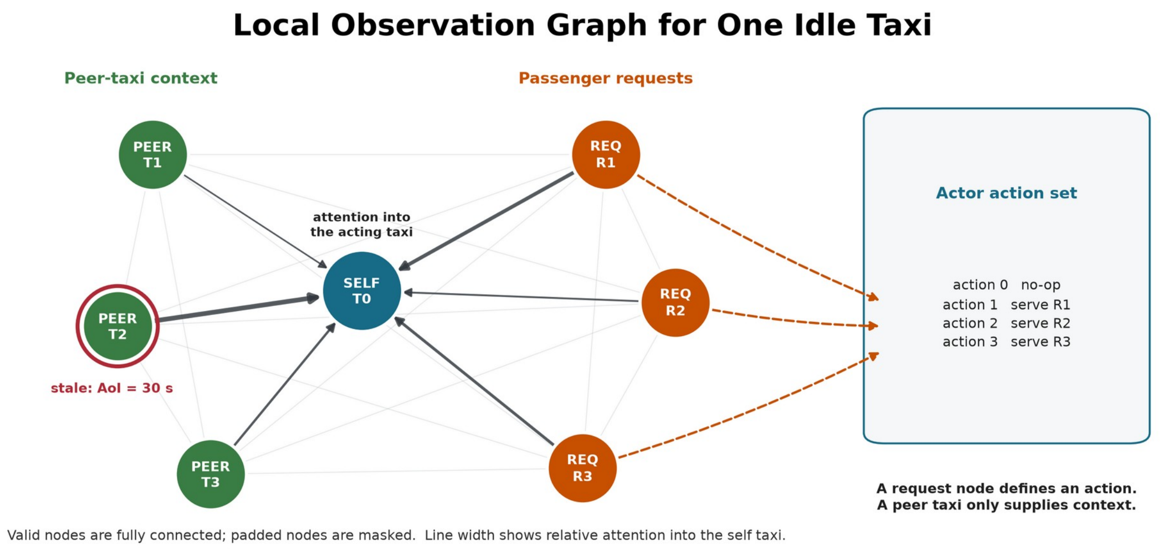

Figure 3.1. Valid self, peer-taxi and request nodes form a fully connected local graph. The red ring marks a stale peer observation, and the illustrative edge widths show attention into the acting taxi. Only request nodes map directly to dispatch actions, which motivates the construct-validity controls.

3.4 Policy models and training

The main policy uses per-node-type feature projections followed by two graph attention layers with four heads. The actor scores no-op and the visible requests. The critic estimates value for MAPPO training with centralized training and decentralized execution. The GAT implementation returns its attention tensors directly so they can be audited without changing the trained network.

The implementation uses scaled dot-product graph attention on the fully connected set of valid local nodes. This differs from the additive attention score in the original GAT paper, so "GAT" in the experiment labels denotes the graph-attention policy family, not an exact reproduction of that layer. For node i, node j, layer l, and head h:

$$\begin{gathered} q_i = W_{Q,h}\,\mathrm{LayerNorm}(h_i^l) \\ k_j = W_{K,h}\,\mathrm{LayerNorm}(h_j^l) \\ v_j = W_{V,h}\,\mathrm{LayerNorm}(h_j^l) \\ s_{ij}^{l,h} = \frac{q_i k_j}{\sqrt{d_{\mathrm{head}}}} \\ \alpha_{ij}^{l,h} = \mathrm{softmax}_j(s_{ij}^{l,h}) \\ z_i = \mathrm{concat}_h(\Sigma_j(\alpha_{ij}^{l,h} \times v_j)) \\ u_i = h_i^l + W_O z_i \\ h_i^{l+1} = u_i + \mathrm{FFN}(\mathrm{LayerNorm}(u_i)) \end{gathered} \tag{3.1}$$

Here, FFN denotes the feed-forward network within each attention layer. The mask removes padded nodes before the softmax. A residual connection keeps the previous node embedding. The actor reads the updated self embedding for no-op and each updated request embedding for its matching dispatch action. The default explanation is the self row (i = 0) averaged over both layers and all four heads, then renormalized:

$$\alpha_j = \text{normalize}\left(\frac{\Sigma_{l,h}\left(\alpha_{0j}^{l,h}\right)}{LH}\right) \tag{3.2}$$

This aggregation is declared before evaluation. A sensitivity analysis also reports each layer, head-wise values, head maximums, and attention rollout; it does not select an aggregation after seeing which result is favorable. The explanation average is an analysis choice: inside the policy, head outputs are concatenated and projected rather than averaged.

The fixed self row represents the acting taxi's dashboard view and is the explanation channel originally declared for audit. It directly contributes to the no-op embedding. A request-action logit, however, is read from that request's updated embedding, so its corresponding request row is more directly aligned with the selected logit. A sensitivity analysis therefore reports a separate request-action sensitivity comparison between the fixed self row and the selected request row. This comparison is not used to redefine the primary metric after seeing the result.

Figure 3.2 summarizes the policy path and the read-only audit channel.

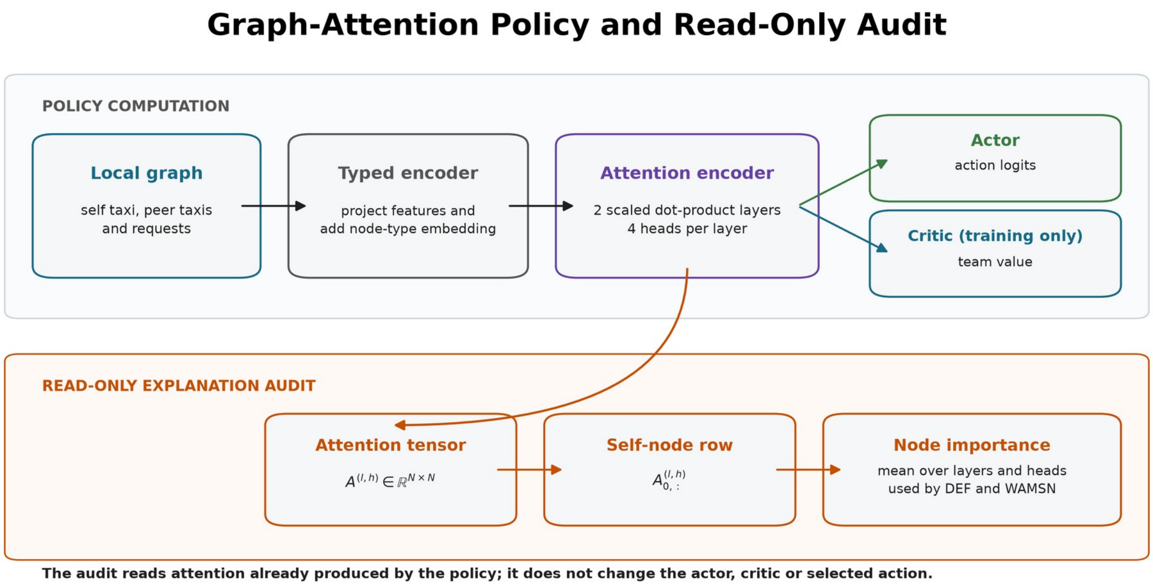

Figure 3.2. The policy and audit share the same two-layer attention encoder. The default audit reduces the self-node attention row to node importance without modifying the actor, critic, or selected action; the selected-request row is evaluated separately as a sensitivity check.

Table 3.2 lists the three policy conditions included in the experiment.

Table 3.2: Policy conditions used in the experiment

Model	Policy	Training observations	Purpose
MLP	MLP-MAPPO	clean	performance context without graph attention
GAT	GAT-MAPPO	clean	primary attention model under audit
GAT-Outage	GAT-MAPPO	tunnel-triggered 30-second outages	degradation-aware training condition

MLP is trained for 50 epochs, GAT for 40 epochs, and GAT-Outage for 50 epochs. Each condition uses independent training seeds 42, 43, and 44.

Candidate checkpoints include periodic snapshots saved every ten epochs, the final snapshot, and a rolling-best snapshot updated when the ten-epoch mean pickup count improves. The implementation records zero-based epoch indices, so index 39 denotes the checkpoint saved after 40 training epochs. Selection uses eight stochastic validation episodes with seed 2026 and mean pickups as the primary criterion; mean reward and the earlier checkpoint index break ties. The test split is not read during selection. Table 3.3 reports the selected checkpoint index for each training run.

Table 3.3: Validation-selected checkpoint indices (zero-based)

Model	seed 42	seed 43	seed 44
MLP	20	40	34
GAT	39	39	30
GAT-Outage	49	40	40

Table 3.4 reports the shared optimization settings.

Table 3.4: Shared training and optimization settings

Setting	Value
Training epochs	40 (GAT), 50 (MLP and GAT-Outage)
Learning rate	0.0001 (GAT variants), 0.0003 (MLP)
Discount factor (gamma)	0.99
Generalized advantage estimation factor (lambda)	0.95
PPO clip ratio	0.20
PPO passes per update	4
Minibatch size	256
Value-loss coefficient	0.25
Value clip ratio	0.20
Entropy coefficient	0.05, adaptive to a 0.20 entropy floor
Maximum adaptive entropy coefficient	0.20
Target Kullback-Leibler divergence	0.015
Gradient-norm limit	0.50
Dispatch credit reward	1.00
Critic encoder gradient scale	0.10
GAT hidden width / layers / heads	64 / 2 / 4
Stability window	Final 10 training epochs
Minimum final-window mean pickups	5.0
Maximum consecutive zero-pickup epochs	4
Minimum selected validation pickups	5.0
Minimum final validation retention	0.80

Final validation retention is the final checkpoint's mean validation pickups divided by the selected checkpoint's mean validation pickups. These four stability gates are applied to every training run before its checkpoint is accepted for evaluation.

At simulation step t, the shared team reward is:

$$r_t = 10\,N_{\text{pickup},t} + 0.5\,N_{\text{dispatch},t} - 0.001\,\text{mean\_wait}_t \tag{3.3}$$

The reward trains dispatch behavior. It contains no term that rewards a faithful, stable, or human-readable attention map.

Figure 3.3 shows how training, validation selection, and held-out evaluation remain separated.

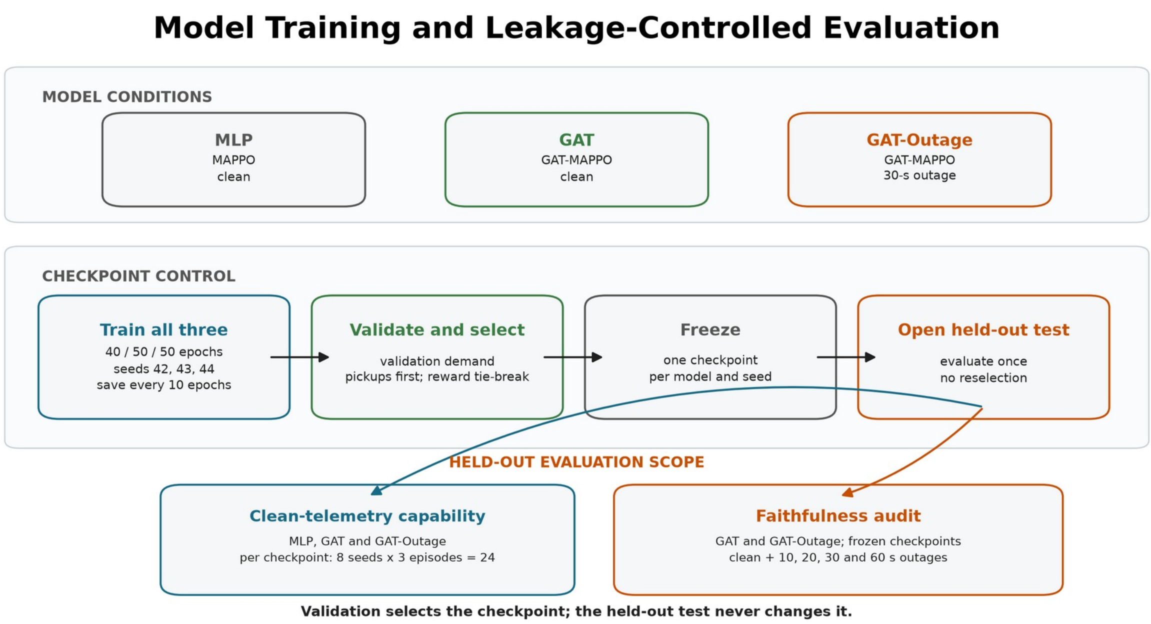

Figure 3.3. Common training and validation-based checkpoint selection for all three model conditions. Each frozen checkpoint receives 24 clean held-out episodes. The faithfulness audit then compares the frozen GAT and GAT-Outage checkpoints across clean and four outage conditions without reselection.

3.5 Telemetry degradation

The trigger and the degradation location are separate parts of the design.

Step 1 - Trigger: when a taxi enters a tunnel edge, the event starts one outage.

Step 2 - Observation-layer outage: for the declared duration, the policy sees the taxi's last valid position and speed. SUMO continues to simulate the true state.

Step 3 - Recovery: after the fixed window expires, a current reading is accepted. A taxi must leave and enter a tunnel again before another tunnel-entry event can trigger.

The fixed outage durations are 10, 20, 30, and 60 seconds. They are easy to monitor and compare, and they create increasing empirical degradation rates. They are not treated as direct causal doses of explanation faithfulness.

AoI is calculated as the current simulation time minus the time of the last valid update and is included in each GAT observation. WAMSN uses normalized AoI as a staleness weight. For this reason, an increase in WAMSN with outage duration partly verifies that the manipulation created longer stale exposure; it is not by itself proof that AoI changed DEF. Figure 3.4 illustrates this separation between physical state and policy observation.

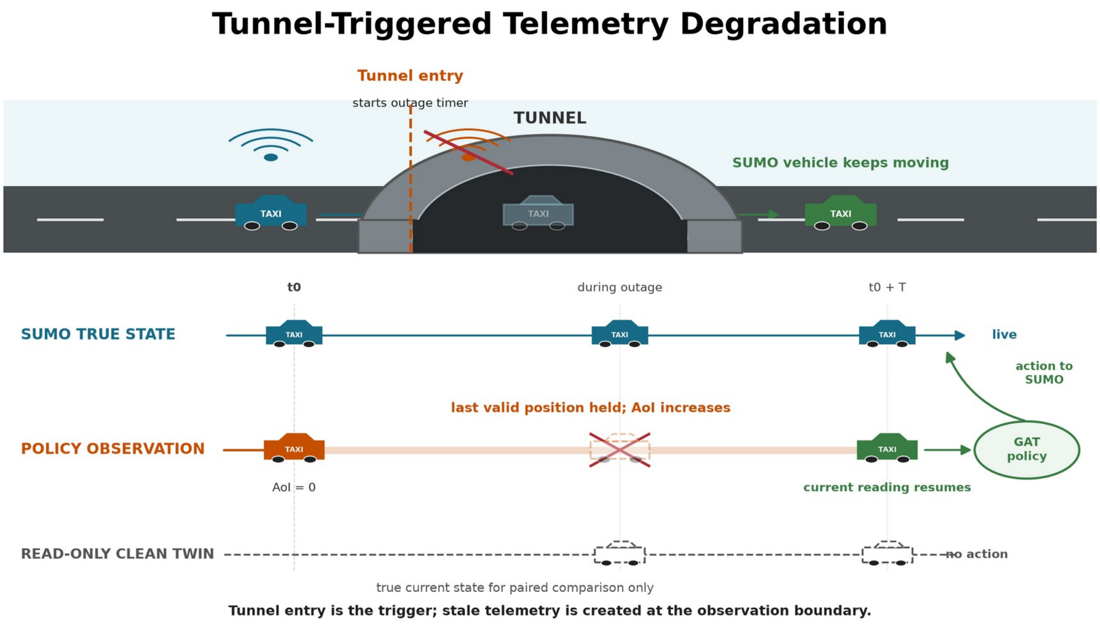

Figure 3.4. Tunnel entry triggers an observation-layer outage. SUMO keeps moving the vehicle while the policy sees its last valid position; the clean twin reads current state but never acts.

3.6 Explanation measures

3.6.1 Decision-level explanation faithfulness

DEF asks whether the nodes ranked highly by an explanation affect the chosen action more than a size- and type-matched random set. Two score functions are recorded from the same counterfactual forward passes. **Probability DEF** uses the selected action probability, p(G,a), and is bounded by the probability scale. **Logit-margin DEF** uses the selected action logit minus the strongest available alternative, m(G,a). The margin can retain resolution when the softmax probability is saturated, but its value depends on a checkpoint's logit scale.

For top-k nodes R_k, comprehensiveness measures the output change when those nodes are removed; sufficiency measures the output retained when only those nodes remain. The two gains are compared with five matched random subsets for each k in {1,2,3} and then averaged.

Let s(G,a) denote either p(G,a) or m(G,a). For explanation subset R_k and a matched random subset B_k:

$$
\begin{aligned}
\mathrm{Comp}(R_k) &= \mathrm{s}(G,a) - \mathrm{s}(G without\ R_k, a)\\
\mathrm{Suff}(R_k) &= \mathrm{s}(G,a) - \mathrm{s}(G keeping\ \ only\ R_k, a)\\
g_{\mathrm{comp}}(k) &= \mathrm{Comp}(R_k) - \mathrm{mean}_b \mathrm{Comp}(B_k^b)\\
g_{\mathrm{suff}}(k) &= \mathrm{mean}_b \mathrm{Suff}(B_k^b) - \mathrm{Suff}(R_k)\\
\mathrm{DEF} &= \mathrm{mean}_k \frac{1}{2}[g_{\mathrm{comp}}(k) + g_{\mathrm{suff}}(k)]
\end{aligned}
\tag{3.4}
$$

Comprehensiveness is better when removing the explanation weakens the action more than removing random nodes. Sufficiency is better when keeping the explanation loses less evidence than keeping random nodes. The logit margin is clipped to [-10, 10] only when an action becomes unavailable or unopposed; every clamp is logged.

Type-matched random subsets are the **control baseline**; they are not themselves a faithfulness score. Corrected DEF is the measured difference between the attention-ranked subset and this matched random control. Its sign is interpreted as follows:

1. **Positive DEF:** the attention-ranked nodes affect the selected action more than comparable random nodes. This supports decision-level faithfulness.
2. **DEF near zero:** the attention ranking has no measured advantage over the matched random control.
3. **Negative DEF:** the attention-ranked nodes affect the selected action less than the matched random nodes. This does not support the attention ranking as a faithful explanation.

These interpretations apply to both variants. The confirmatory H1-H4 family and the paired clean-twin comparison use probability DEF, following the declared hypothesis analysis. Logit-margin DEF is used for the construct-validity and ranking-control diagnostics because it can show output movement that a saturated probability hides. It is action-protected where request deletion could remove the chosen action. The two variants have different units and cannot be converted into one another. Both use the type-matched baseline described in Section 3.7. DEF provides comparative perturbation evidence; even a positive value does not prove that attention is a complete causal explanation.

The study uses related measures for different questions. They should not be read as interchangeable estimates. Table 3.5 states the role of each measure.

Table 3.5: Explanation measures and roles

Measure	**Decisions and score**	**Role in the study**
Probability DEF	all scorable sampled actions; probability scale	primary H1-H4 and paired clean/degraded analysis
Logit-margin DEF	all scorable sampled actions; selected-action margin	construct-validity and ranking diagnostics when probability is saturated
Dispatch self-row margin-DEF	sampled dispatch actions only; declared self-row explanatio n	release check for decision relevance under clean and 60-second outage conditions

Measure	Decisions and score	Role in the study
WAMSN	stale-exposed decisions; AoI-weighted attention	freshness-exposure indicator and H3 manipulation check
Paired attention and DEF shifts	degraded observation minus its clean twin	direct within-decision response to the observation-layer intervention

3.6.2 Stale-node attention

WAMSN measures attention attached to stale vehicle information:

$$\mathrm{WAMSN} = \frac{\Sigma_i\left(\mathrm{attention}_i \times \mathrm{normalized\_AoI}_i\right)}{\Sigma_i\left(\mathrm{attention}_i\right)} \tag{3.5}$$

Only self and peer-taxi nodes contribute because request nodes do not carry vehicle telemetry. The analysis reports WAMSN when at least one stale vehicle node is visible, together with stale-node attention share, stale-node mass, and whether a stale node enters the top three.

The paired stale-attention shift compares the degraded observation with its exact clean twin at the same simulation step. Positive values mean the degraded observation assigns more attention mass to nodes marked stale; negative values mean it assigns less.

3.7 Construct-validity audit

The audit supports the primary research problem by checking whether DEF is a valid comparison in this action-candidate graph.

A uniform random occlusion can select passenger requests more often than the attention top-k. Removing such a request changes both the observation and the available action set. A nearby-taxi occlusion changes only information. The result can therefore reflect different node-type composition rather than explanation quality.

The corrected protocol uses three controls:

- **type matching:** random subsets contain the same mixture of taxi and request nodes as the explanation subset;
- **chosen-action protection:** the request associated with the selected action is not deleted in the action-protected variant;
- **clamp logging:** counterfactual action-margin clamps are counted so that action deletion can be detected.

Figure 3.5 illustrates why these controls are required.

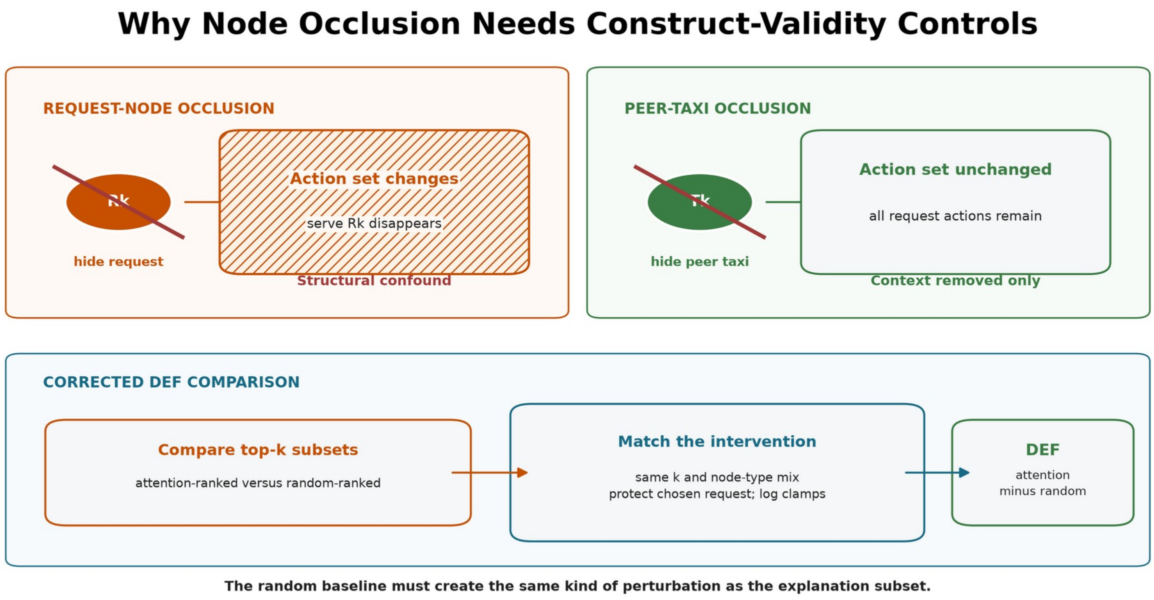

Figure 3.5. Request-node occlusion can remove its action, while peer-taxi occlusion removes context only. Corrected DEF matches subset size and node types, protects the chosen request, and logs margin clamps.

Uniform-baseline results are retained only as an audit trail. They are not used for the final hypothesis verdicts.

Three additional diagnostics test whether a near-zero DEF can be interpreted. First, a leave-one-out (LOO) control masks each valid node separately and ranks nodes by the resulting selected-action margin loss. The selected request is protected. This control is expected to strengthen comprehensiveness because it is built from the same single-node perturbation, but it is not an independent proof that the combined DEF is optimal. Second, Gradient x Input ranks each node by the L2 norm of the selected-action logit gradient multiplied by its raw features. It is a post-hoc comparator that does not use attention weights. Third, the audit reports valid taxi/request counts and the exact expected top-k overlap with a type-matched random subset. Taxi-only Spearman correlation between attention and LOO effects supplies a ranking measure that does not delete request actions and does not depend on random subset overlap.

3.8 Evaluation matrix

MLP, GAT, and GAT-Outage are evaluated under clean telemetry with eight evaluation seeds (42-49) and three episodes per seed. One checkpoint is frozen for each model and training seed, so this gives 24 held-out episodes per checkpoint. The clean capability evaluation therefore contains 216 episode evaluations:

$$3 \times 3 \times 8 \times 3 = 216 \tag{3.6}$$

Table 3.6 summarizes the complete evaluation structure. An episode evaluation is one complete rollout of one frozen checkpoint under one telemetry condition.

Table 3.6: Held-out evaluation structure and episode counts

Evaluation	Model families	Total frozen checkpoints	Telemetry cases	Episodes/cell	Total
Clean capability context	3	9	1	24	216

Evaluation	Model families	Total frozen checkpoints	Telemetry cases	Episodes/cell	Total
Exposure-conditioned faithfulness sweep	2	6	5	48	1,440
Added ranking controls	2	6	2	24	288
Random-trigger sensitivity analysis	2	6	3	24	432

GAT and GAT-Outage receive the full faithfulness sweep:

$$2 \times 3 \times 5 \times 8 \times 6 = 1{,}440 \tag{3.7}$$

Table 3.7 lists the five telemetry conditions and their roles.

Table 3.7: Telemetry conditions used in held-out evaluation

Telemetry condition	Observation-layer treatment	GAT-Outage relation	Evaluation purpose
Clean	Current vehicle readings	No imposed outage	Capability context and faithfulness baseline
10 s	Freeze last valid reading for 10 seconds	Unseen shorter duration	Mild-degradation transfer test
20 s	Freeze last valid reading for 20 seconds	Unseen shorter duration	Intermediate transfer test
30 s	Freeze last valid reading for 30 seconds	Training-matched duration	In-condition degradation test
60 s	Freeze last valid reading for 60 seconds	Unseen longer duration	Severe stress and transfer test

The legal-random capability baseline samples uniformly from no-op and the currently valid request actions. The greedy baseline is deliberately naive: each idle taxi chooses its nearest visible request independently. If several taxis choose the same request, the environment accepts the first action in the step and rejects the later conflicts; there is no fleet-wide assignment or conflict optimization. The policy is deterministic for a fixed demand variant, so its uncertainty is based on three independent demand variants rather than the repeated evaluation-seed records. It is therefore interpreted only as a lower bound, not as a competitive dispatch algorithm.

The primary learned-policy evaluation samples from the policy distribution. This matches training and preserves both no-op and request-dispatch decisions for explanation auditing. A separate descriptive diagnostic evaluates each of the six selected GAT checkpoints by deterministic argmax over three held-out test episodes. This diagnostic is not used for checkpoint selection or the primary capability comparison. It checks whether sampled performance also corresponds to a usable deterministic policy.

The relation column refers only to GAT-Outage; the clean-trained GAT has not seen any imposed outage duration during training. The evaluation changes the observation-layer condition but never retrains or reselects a checkpoint. Outside stale exposure, faithfulness is sampled every 16 decisions. Every decision that contains at least one stale vehicle node is scored. Each such observation is also compared with its exact clean twin at the same simulation step. The degraded and clean evaluations reuse the same matched-random draw, so the paired difference does not contain avoidable baseline-sampling noise.

Each degraded condition must contain at least 20 stale-exposed records from at least three episodes. Preflight also checks expected cells, clean source provenance, type-matched controls, chosen-action exclusion, complete event capture, and separation in empirical AoI between conditions. All six sweeps pass these gates.

3.9 Hypotheses and statistics

The confirmatory hypotheses are:

- **H1:** DEF decreases as outage duration increases.
- **H2:** faithfulness declines faster than dispatch performance.
- **H3:** WAMSN increases as outage duration increases.
- **H4:** decisions with greater WAMSN have lower DEF within an episode.
- **H5:** compared with clean training, degradation-aware training reduces the decline in DEF associated with longer outages and higher WAMSN.

H3 is interpreted as a stale-exposure manipulation check because WAMSN includes normalized AoI by definition. It confirms that longer outages create more AoI-weighted exposure; it is not independent evidence that AoI changes DEF.

H1 and H3 use episode-block permutation trend tests and cluster bootstrap confidence intervals. H2 uses paired cell-level sign flips relative to each evaluation seed's clean condition. H4 calculates Spearman correlation within each episode before a sign-flip test. Holm correction is applied to H1-H4 within each trained policy.

H5 uses a matched-seed directional comparison rather than a cross-seed significance test. For training seed s, H5 is supported only if the GAT-Outage checkpoint has both an H1 duration-DEF correlation and an H4 WAMSN-DEF correlation closer to zero than the clean-trained GAT checkpoint with the same seed. In other words, both adverse negative relationships must be weaker after degradation-aware training. If either relationship is equally or more negative, that seed does not support H5. The result is reported as the number of supporting matched seeds out of three; no population p-value is claimed from these three comparisons.

The exposure-conditioned paired follow-up directly subtracts clean-twin DEF from degraded DEF for the same decision. Negative values mean lower measured faithfulness under the degraded observation. This paired estimate is reported with an episode-block confidence interval and is interpreted by effect size, direction across trained policies, and uncertainty rather than by a decision-level p-value alone.

H2 is treated as a predeclared exploratory comparison. Its original rate divides by clean DEF, which is close to zero and can make a small absolute change look arbitrarily large. The analysis therefore reports the support verdict and absolute trajectories, but does not use the ratio as the main evidence for faithfulness decoupling.

Thousands of decisions improve measurement precision but do not replace independent model replication. The final synthesis therefore treats each training seed as one trained-policy replicate. With only three seeds per model, the dissertation reports the range and number of supporting seeds and does not claim a cross-seed population p-value.

An exploratory action-stratified audit separates scorable decisions with at least one available request into no-op and dispatch. Decisions with no available request are excluded because no-op is then forced rather than chosen. Within each stratum, records are still averaged by episode before permutation tests or bootstrap intervals are formed. This diagnostic tests whether the combined statistics describe both actions or are driven by the more common action. It was added after the primary analysis and is interpreted descriptively rather than as a new confirmatory hypothesis.

A second sensitivity analysis checks whether the paired result is specific to the fixed tunnel location. The six checkpoints remain frozen and are evaluated under clean telemetry, a 30-second tunnel-triggered freeze, and a 30-second randomly triggered freeze. The random trigger is an independent Bernoulli draw for each taxi at each observation step. A preliminary run without faithfulness scoring selected a trigger probability of 0.0023: for the reference GAT checkpoint, this produced a 0.569% degraded-observation rate, close to the 0.580% tunnel rate. The same probability is then held fixed for all checkpoints. Because policy actions change taxi availability and trajectories, the realized exposure rates can differ; they are reported rather than assumed equal. The comparison is therefore interpreted mainly from paired shifts among decisions that actually contain a stale vehicle node.

The ranking controls use the same held-out demand and stochastic action protocol. They evaluate GAT and GAT-Outage under clean telemetry and a 60-second outage:

$$2 \times 3 \times 2 \times 8 \times 3 = 288 \tag{3.8}$$

LOO, Gradient x Input, overlap, and aggregation diagnostics are sampled every eight decisions. Query-row sensitivity is evaluated at every selected-request action because it asks whether the explanation row matches the node that produces that action logit. Decision records are averaged within episodes before bootstrap confidence intervals are calculated. Attention-aggregation sensitivity is reported per training seed and only for decisions where at least one stale vehicle node is visible.

3.10 Audit decision implementation

The experiment pipeline produces measurements; a separate release-audit stage turns those measurements into a decision about whether attention may be shown as an explanation. The stage reads saved evidence and leaves the policy, selected action, SUMO state, and observations unchanged.

Its inputs are the frozen-checkpoint preflight reports, condition summaries, action-stratified results, training-seed synthesis, deterministic diagnostics, faithfulness controls, and tunnel-versus-random trigger comparison. These artifacts jointly cover telemetry freshness, decision relevance, action composition, attention extraction, checkpoint replication, and the action rule intended for deployment.

The implementation applies nine checks:

1. evidence integrity;
2. separate freshness reporting;
3. type-matched and action-protected controls;
4. dispatch-action decision relevance;

5. no-op and dispatch composition;
6. attention-extraction stability;
7. consistency across independently trained checkpoints;
8. capability of the action rule intended for deployment; and
9. robustness to tunnel and random loss triggers.

Decision relevance passes only when the 95% confidence-interval lower bound of self-row dispatch margin-DEF is greater than zero under both clean and 60-second outage evaluation. A positive value means that the attention ranking is more informative than its type-matched random control. Extraction stability fails if an audited layer, head, maximum, or rollout rule reverses the direction produced by the declared self-row mean. Trigger robustness uses confidence intervals: a direction is supported only when its 95% interval excludes zero, and supported tunnel and random-trigger directions must agree. An interval that crosses zero is INDETERMINATE, not evidence of a reversal. The remaining rules test evidence availability, action coverage, checkpoint agreement, and the action rule declared for deployment.

The present audit concerns actions sampled from the policy distribution. The argmax evaluation is therefore retained as a descriptive limitation rather than a mandatory release gate. If an argmax policy were intended for deployment, its held-out capability check would become mandatory. The current action-composition gate confirms that both action strata and dispatch evidence are present; it does not claim that a sparse dispatch stratum would be representative.

The output has three states. ELIGIBLE means that every required check passed within the stated scope. WITHHOLD means that at least one check failed and attention must remain an internal diagnostic. INCOMPLETE means that required evidence is missing, so no release decision can be made. Eligibility is not proof of a complete causal explanation; it only permits presentation with an explicit freshness indicator and audit scope as an audited candidate explanation.

The release rules are a post-study synthesis of the risks identified by the experiment. Their application to the present results is therefore demonstrative and is not an additional confirmatory test of H1-H5. For future candidate models, the rules and thresholds must be fixed before held-out evaluation.

3.11 Reproducibility

The experiment runner records the configuration, commands, seeds, checkpoint hashes, source revision, and preflight reports in versioned run directories. Compact outputs include the summary, performance context, deterministic diagnostic, action-stratified audit, and training-seed synthesis tables. The training-seed synthesis makes consistency across trained policies explicit. Reproduction commands, file locations, and expected outputs are documented in docs/REPRODUCE_EXPERIMENTS.md. The complete frozen-checkpoint workflow is available as a single framework stage. It produces a machine-readable JSON decision, a CSV check table, and a reader-facing Markdown report. The detailed input and output contract is documented in docs/FRESHNESS_AWARE_EXPLANATION_AUDIT.md.

3.12 Ethics and data governance

The experiment uses a simulated taxi fleet, generated demand, and an OpenStreetMap-derived road network. It contains no human participants, personal records, live vehicle identifiers, or commercial dispatch data. Its main ethical risk is miscommunication: presenting attention as a trustworthy reason could give an operator false confidence. The reporting therefore keeps freshness, task behavior, and faithfulness separate and avoids unsupported causal claims.

Chapter 4: Results

The automated preflight audit passed 100 checks covering provenance and protocol requirements. Results are reported by training seed because decisions and episodes from one checkpoint do not constitute independent model replications.

4.1 Policy capability and training stability

Validation selected GAT checkpoint indices 39, 39, and 30 and GAT-Outage indices 49, 40, and 40 for seeds 42, 43, and 44. These are zero-based indices; for example, index 39 is the checkpoint saved after 40 training epochs. The selected index is reported separately for each trained policy.

Figure 4.1 shows the training and checkpoint-selection diagnostics.

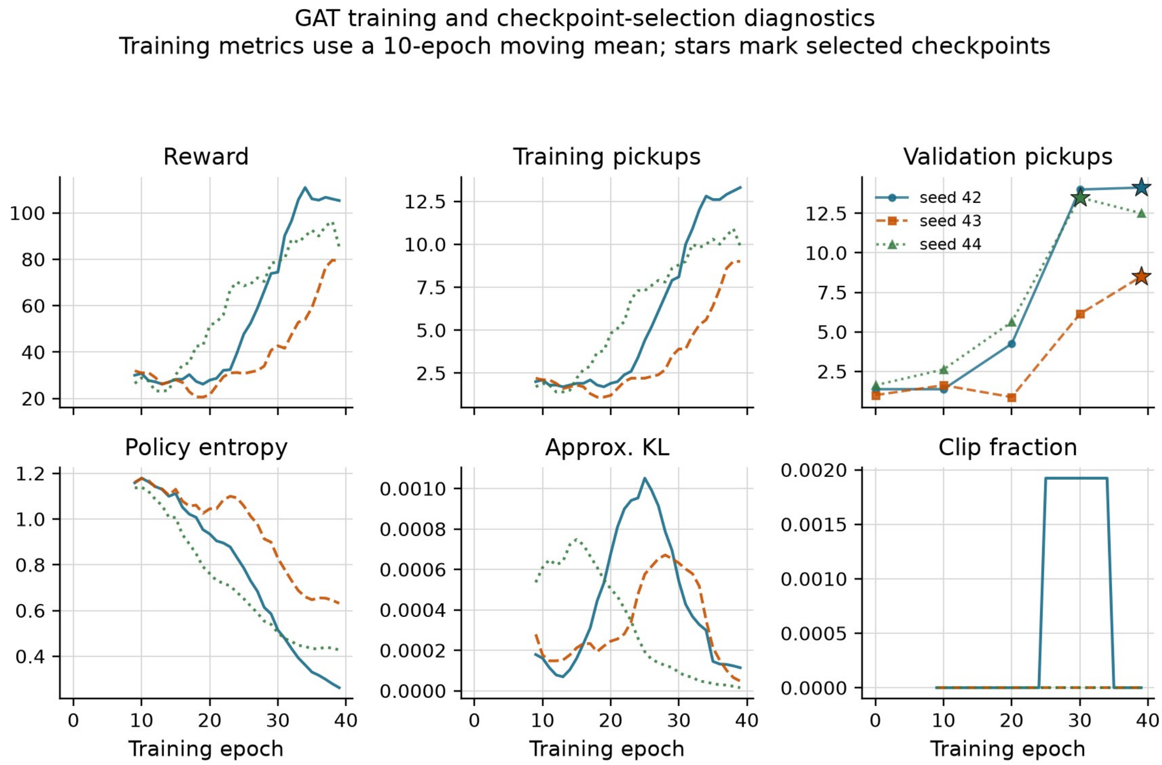

Figure 4.1. GAT training metrics use a ten-epoch moving mean. Stars mark validation-selected checkpoints. Reward and pickups still vary between epochs, but entropy remains above the collapse threshold and PPO clipping is limited.

All nine training runs pass the declared stability gates. The following values describe the final ten training epochs, not the held-out evaluation in Table 4.1. For GAT, the final ten-epoch mean pickups are 13.3, 9.0, and 9.9, and final validation retention is 100%, 100%, and 92.6%. The corresponding GAT-Outage values are 12.4, 14.1, and 13.8 pickups, with validation retention of 100%, 93.9%, and 96.5%. Training metrics still fluctuate across epochs, but no selected checkpoint is a zero-pickup policy under stochastic validation. Deterministic argmax results are reported separately below.

Table 4.1 places the selected policies above two lower bounds evaluated on the same held-out demand. Performance is reported only to establish that the audited policies perform above random and greedy lower bounds.

Table 4.1: Policy capability on held-out demand

Policy	Mean pickups	Uncertainty or replicate range
Legal random	1.50	95% episode-cluster CI [1.08, 1.96]
Greedy nearest request	2.00	95% demand-variant CI [1.00, 3.00]
MLP	13.63	training-seed range 12.83-14.17
GAT	11.64	training-seed range 9.42-13.71
GAT-Outage	12.97	training-seed range 11.79-14.58

The legal-random baseline has 24 independent episode clusters. Greedy nearest is deterministic, so its effective sample is the three demand variants rather than 24 repeated seed-episode records. GAT's seed means are 13.71, 9.42, and 11.79. Every selected policy exceeds both lower bounds under stochastic action sampling. These results confirm capability for the sampled policy distributions only; the role of graph attention and the usability of the argmax policy remain untested.

The audit requires each checkpoint's sampled policy distribution to exceed the declared lower bounds, not GAT to outperform MLP. That condition is met. MLP has the highest cross-seed mean in this implementation, although the replicate ranges overlap. The experiment finds no task-performance advantage for graph attention. This limits how broadly the audit can be generalized but remains separate from whether the exposed weights faithfully explain GAT decisions.

Figure 4.2 shows the held-out pickup count for each selected policy.

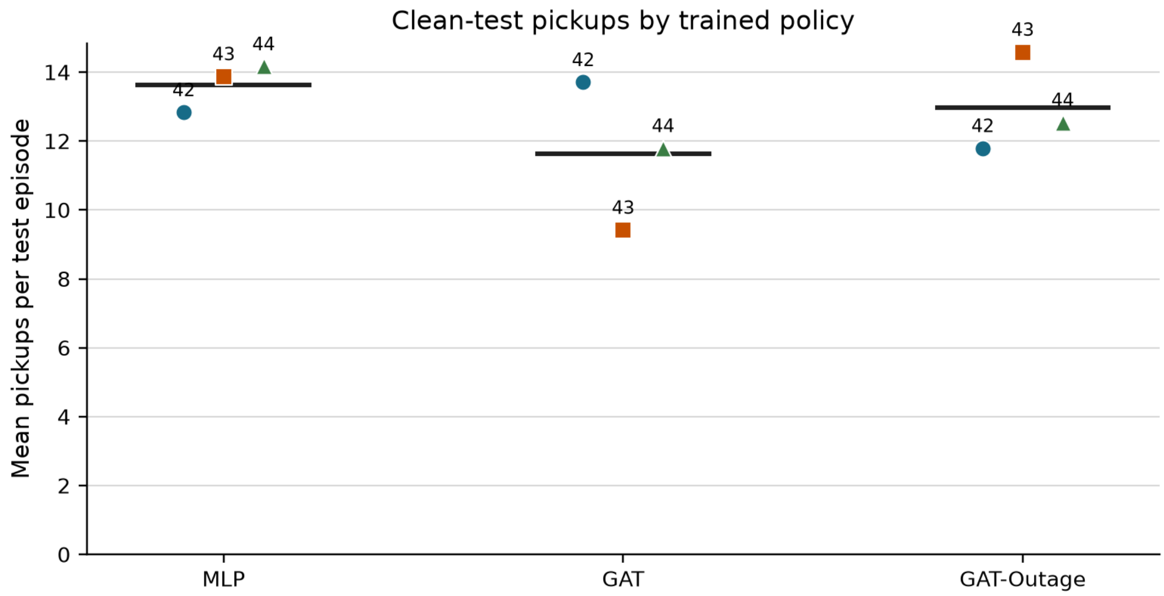

Figure 4.2. Each point is one selected policy evaluated over 24 held-out episodes. Horizontal lines show the cross-seed mean.

The deterministic diagnostic changes the capability interpretation. All six selected graph-attention checkpoints choose no-op throughout three held-out episodes, giving 0 pickups in 18 of 18 checkpoint-episodes. Their common mean reward is -46.58 and final pending-request wait is 905.9 seconds. The stochastic results show only that the policy distributions assign enough probability to dispatch actions to produce pickups when sampled; they do not establish a stable argmax dispatcher.

4.2 Construct-validity audit

Removing a request node can remove an available action, while removing a peer taxi only hides information. A uniform random occlusion therefore compares different types of change. Table 4.2 quantifies the resulting control-baseline artifact.

Table 4.2: Construct-validity audit of random controls

Control baseline used to calculate DEF	Mean margin-DEF	95% CI	Interpretation
Uniform random	-0.5540	[-0.5808, -0.5288]	dominated by node-type and action-deletion imbalance
Type-matched random	-0.0093	[-0.0159, -0.0013]	small negative result after matching node types

The type-matched interval excludes zero, but its magnitude is about two orders smaller than the uniform-control artifact. This 1,199-decision audit supports the measurement design rather than answering the primary research question. The reported protocol uses type matching, protects the chosen request in the margin variant, and logs action-margin clamps.

4.3 Evaluator sensitivity and ranking controls

Raw attention, Gradient x Input, and an LOO perturbation ranking are evaluated under clean and 60-second conditions. Each model, training seed, and condition contains eight evaluation seeds and three episodes. Table 4.3 shows clean results. The 60-second aggregate means remain close to clean.

The two conditions are stored as separate evaluations and have different decision counts. Across the six checkpoints, the largest absolute 60-second minus clean change is 0.000071 for raw attention, 0.000254 for Gradient x Input, and 0.000306 for LOO. Figure 4.3 shows the clean values and the condition difference separately instead of presenting two visually indistinguishable rows.

Table 4.3: Clean-telemetry faithfulness controls by training seed

Model	Seed	Raw-attention margin-DEF	GxI margin-DEF	LOO margin-DEF	Taxi-only rho
GAT	42	+0.0003	-0.0055	+0.0008	+0.088
GAT	43	-0.0009	-0.0033	+0.0099	+0.404
GAT	44	+0.0015	+0.0012	+0.0030	+0.426
GAT-Outage	42	+0.0011	-0.0019	+0.0020	+0.472
GAT-Outage	43	+0.0006	-0.0010	+0.0089	+0.602
GAT-Outage	44	+0.0031	+0.0052	+0.0044	+0.699

The LOO-ranked control is positive and larger than raw attention for all six selected checkpoints. This shows that DEF responds to a ranking built from single-node margin loss. LOO is still a perturbation control, not a ground-truth explanation. Gradient x Input changes sign across checkpoints and provides no uniform improvement.

Raw-attention DEF ranges from -0.0009 to +0.0031. Taxi-only rank correlation with LOO is positive in all six checkpoints, but ranges from 0.088 to 0.699. Attention shows some agreement with decision-relevant LOO ordering, but the strength of that agreement remains checkpoint-dependent.

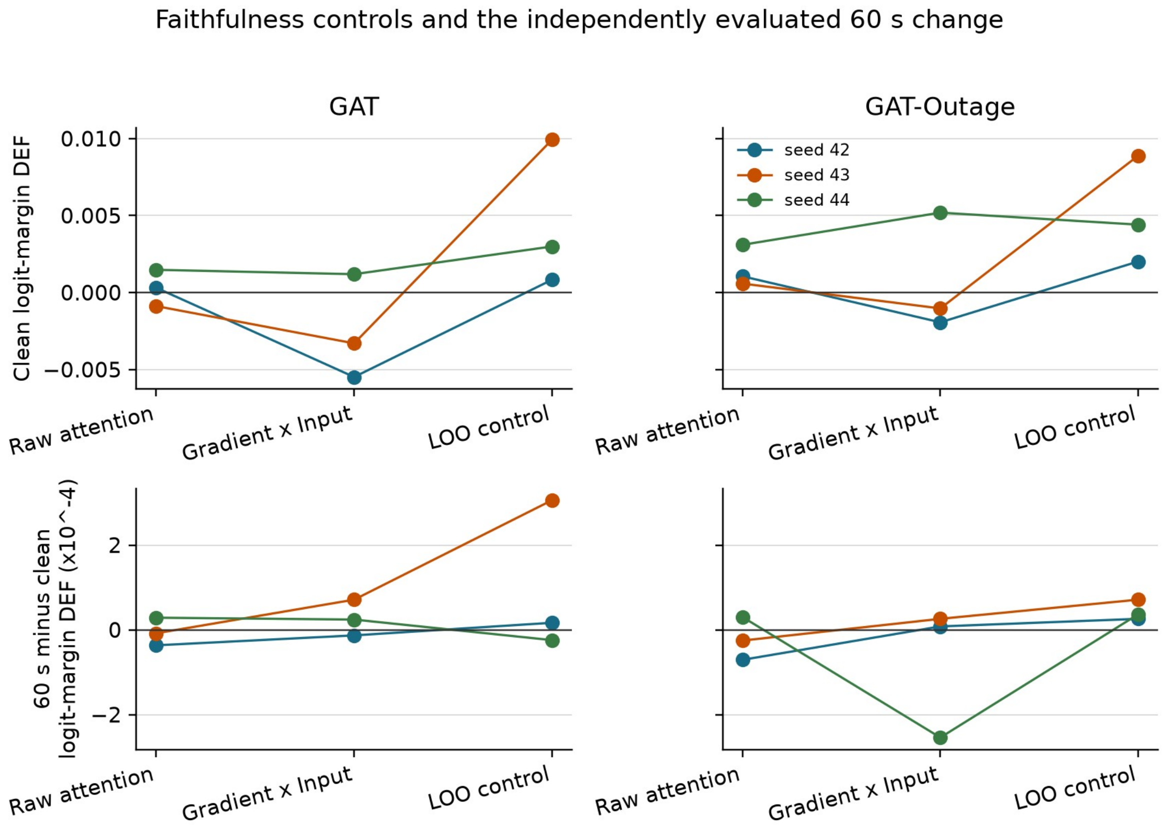

Figure 4.3. Top: clean logit-margin DEF. Bottom: 60-second minus clean DEF, multiplied by 10,000, from separate evaluations. Lines join methods for each checkpoint; non-zero changes confirm distinct records.

The attention reduction rule remains important. Across the declared mean, individual heads, layer maxima, and rollout, every checkpoint contains both a positive and a negative stale-attention shift. The widest ranges are -0.0109 to +0.0063 for GAT seed 44 and -0.0097 to +0.0064 for GAT-Outage seed 44.

Figure 4.4 shows the sensitivity to the attention reduction rule.

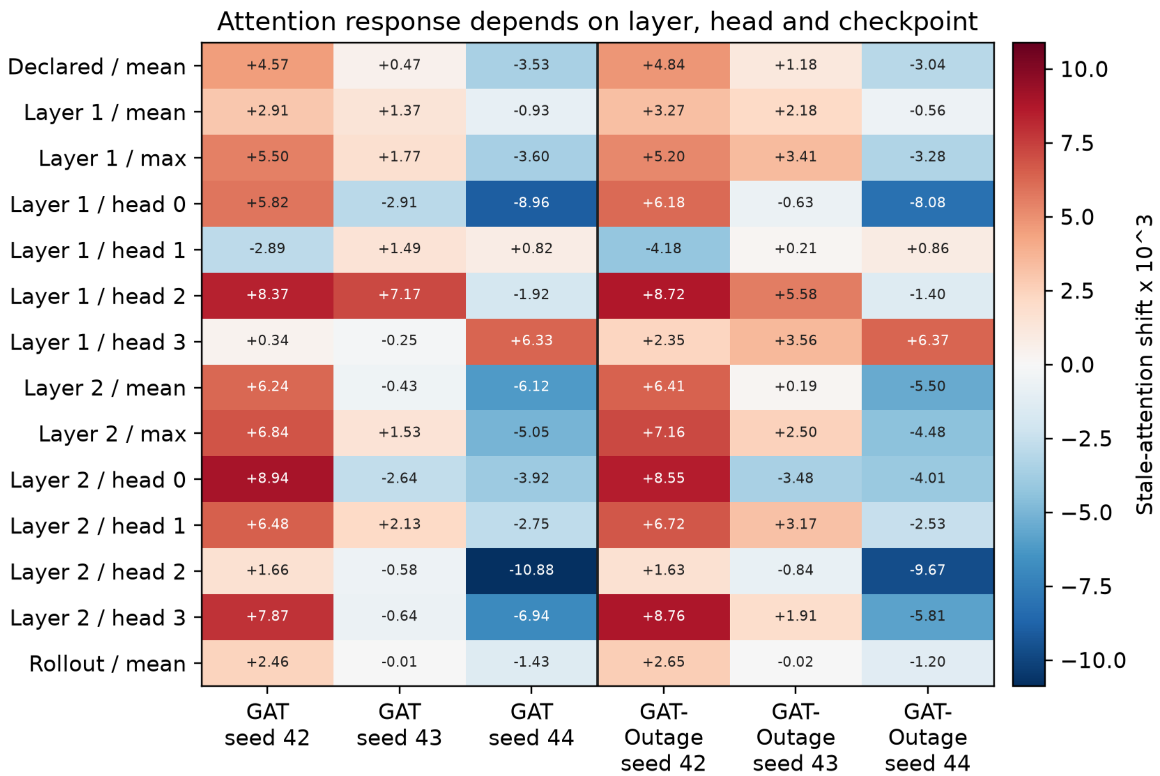

Figure 4.4. Stale-attention shift multiplied by 1,000. Selecting a layer, head, or rollout rule after seeing the result could reverse the conclusion.

Changing from the declared self row to the selected request row has little effect for seeds 42 and 44, but increases request-action DEF by about 0.0064 for GAT seed 43 and 0.0073 for GAT-Outage seed 43. The aggregate gap is +0.0022 for GAT and +0.0025 for GAT-Outage because it averages these different policy responses. The largest absolute 60-second minus clean query-row change is 0.000264, so the two conditions again remain close without being duplicates.

Figure 4.5 compares the declared self row with the selected request row.

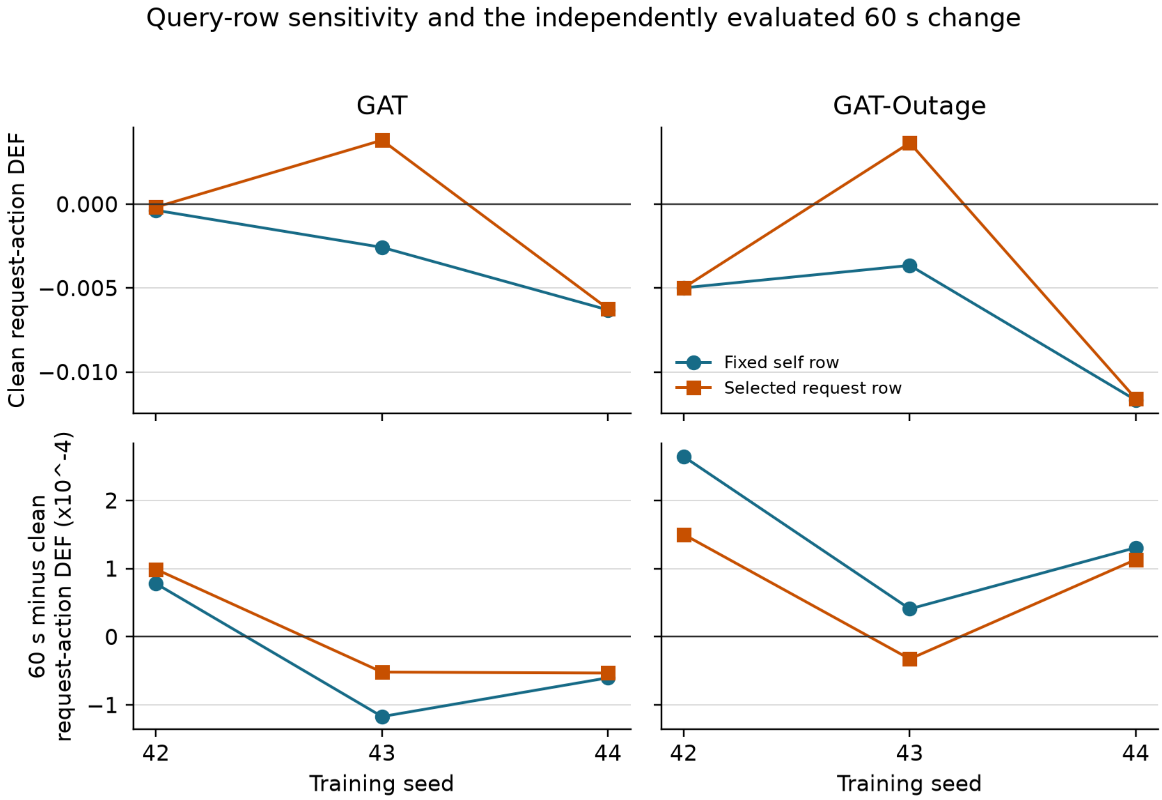

Figure 4.5. Top: clean request-action DEF from the declared self row and the selected request row. Bottom: each query row's independently evaluated 60-second result minus clean, multiplied by 10,000. The selected request row improves the seed-43 clean results but does not provide a consistent improvement across checkpoints or conditions.

4.4 Small-graph resolution

The scored graphs usually contain the maximum number of visible nodes, as summarized in Table 4.4.

Table 4.4: Valid node-count distribution in scored graphs

Model	Node count	Mean	Median	5th percentile	95th percentile
GAT	peer taxis	5.00	5	5	5
GAT	requests	4.51	5	1	5
GAT	non-self total	9.51	10	6	10
GAT-Outage	peer taxis	5.00	5	5	5
GAT-Outage	requests	4.52	5	1	5
GAT-Outage	non-self total	9.52	10	6	10

Even so, type matching creates substantial top-k overlap. The expected random intersection is about 0.22 for k=1, 0.41 for k=2, and 0.60 for k=3. Attention-LOO agreement at k=1 exceeds random overlap for both models. At k=3, it is 0.60 for GAT and 0.49 for GAT-Outage, compared with expected overlap of about 0.60. Restricting analysis to at least six non-self nodes does not change the sample because every scored decision already meets that threshold. This pattern suggests that the ranking agreement has more contrast at k=1; it does not establish that LOO is ground truth or that the top-ranked attention node is a causal explanation.

Figure 4.6 shows how the expected overlap grows with k.

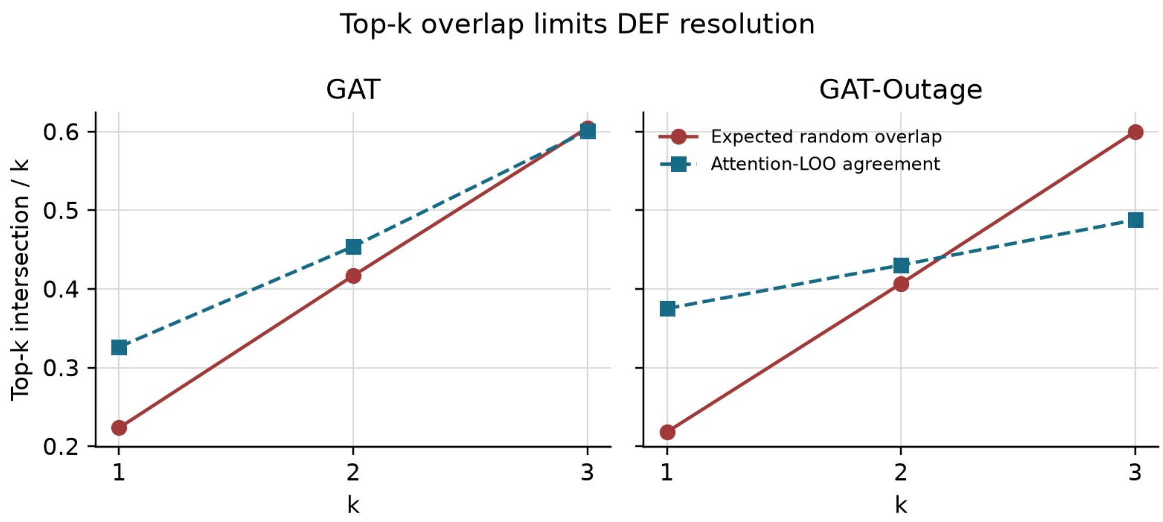

Figure 4.6. Large expected overlap between the explanation and its matched random set reduces the contrast available to DEF, especially at k=3.

4.5 Telemetry manipulation and stale exposure

Tunnel-triggered observation outages create increasing stale exposure. The event-aware audit scores every decision that contains at least one stale taxi, instead of relying on periodic sampling. Across GAT checkpoints, the number of stale-exposed records ranges from 347-382 at 10 seconds and 1,476-1,806 at 60 seconds. The GAT-Outage ranges are 353-372 and 1,566-1,697. Every degraded cell exceeds the pre-declared coverage gates.

Conditional WAMSN rises from 0.0275-0.0289 at 10 seconds to 0.0804-0.0900 at 60 seconds. H3 is supported for all six selected checkpoints. Because WAMSN contains normalized AoI, this verifies increasing stale exposure; it does not show that AoI causes lower faithfulness. Figure 4.7 displays the duration sweep.

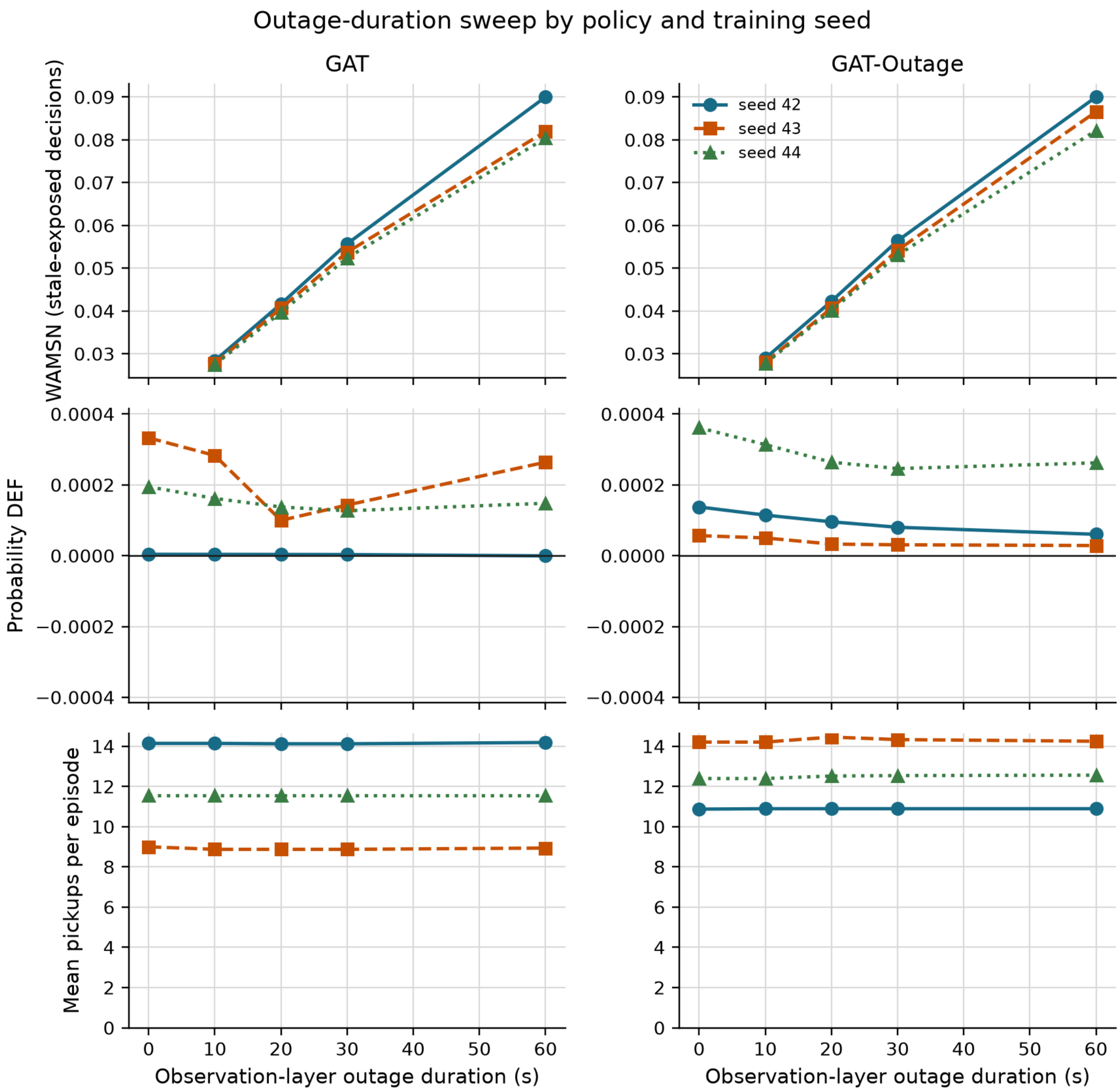

Figure 4.7. Longer observation outages increase WAMSN for every policy. DEF changes are much smaller; pickups are reported only as context, not as the main research outcome.

4.6 Exposure-conditioned paired audit

The paired audit compares each degraded observation with its exact clean twin at the same simulation step. Positive attention shift means more attention is assigned to nodes marked stale. Negative DEF shift means lower measured faithfulness under degradation. Table 4.5 reports the paired estimates.

Table 4.5: Exposure-conditioned paired attention and DEF shifts

Model	Seed	Attention shift x 1,000 [95% CI]	Paired probability-DEF shift x 10,000 [95% CI]
GAT	42	+4.074 [+3.9, +4.2]	-0.041 [-0.060, -0.022]
GAT	43	+0.208 [+0.2, +0.3]	-0.839 [-1.892, +0.143]
GAT	44	-3.483 [-3.6, -3.4]	+0.092 [+0.068, +0.116]
GAT-Outage	42	+3.975 [+3.8, +4.1]	-0.253 [-0.323, -0.193]
GAT-Outage	43	+0.959 [+0.9, +1.0]	-0.114 [-0.343, +0.061]

Model	Seed	Attention shift x 1,000 [95% CI]	Paired probability-DEF shift x 10,000 [95% CI]
GAT-Outage	44	-3.038 [-3.1, -2.9]	+0.172 [+0.135, +0.212]

Figure 4.8 shows the paired attention and DEF shifts by checkpoint.

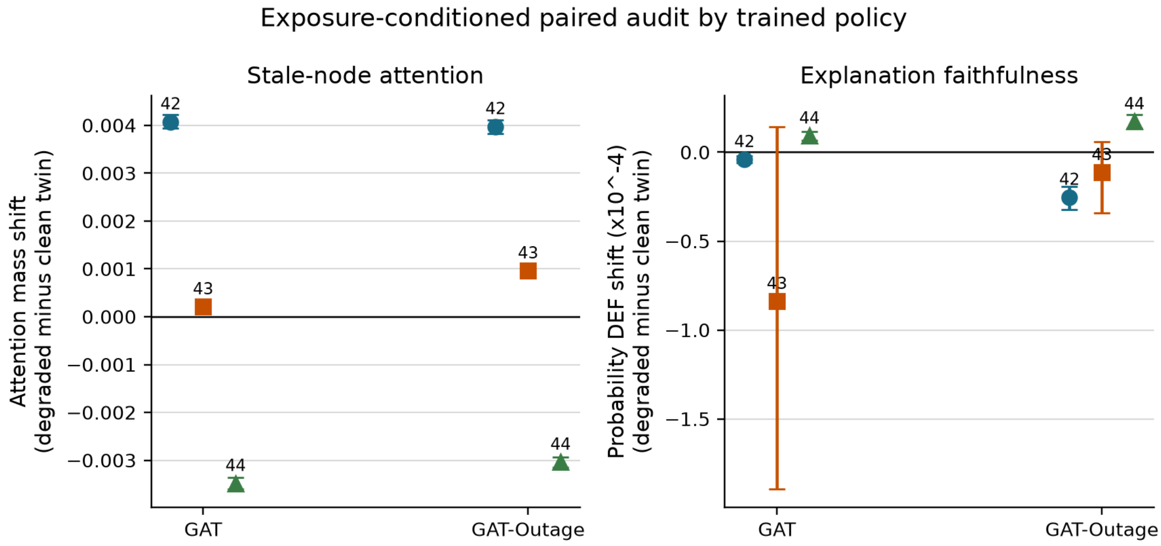

Figure 4.8. Checkpoints trained with seeds 42 and 43 move attention toward stale nodes and have a negative mean DEF shift. Checkpoints trained with seed 44 show the opposite directions. Error bars are 95% episode-block bootstrap intervals.

The same checkpoint pattern appears in both training regimes: four of six selected checkpoints have a positive attention shift and four have a negative mean DEF shift. Only the DEF decrease for checkpoints trained with seed 42 is separated from zero in both models; both seed-43 intervals include zero, while both checkpoints trained with seed 44 show a small increase. The effects are numerically small and directionally heterogeneous. This is not evidence for a universal claim that telemetry degradation moves attention toward stale nodes or always lowers faithfulness. The largest absolute paired probability-DEF shift is 0.0000839, equivalent to 0.00839 probability percentage points. Relative percentages are not reported because clean DEF is close to zero and would make such ratios unstable.

4.7 Action-stratified diagnostic

The combined faithfulness audit is dominated by chosen no-op actions. Across all six checkpoints, 56,550 records contain at least one valid request. Of these, 52,585 (93.0%) select no-op and 3,965 (7.0%) dispatch a request. Decisions with no valid request are excluded from these percentages. Table 4.6 reports the action composition and dispatch-only diagnostic.

Table 4.6: Action composition and dispatch-stratified DEF diagnostic

Checkpoint	No-op (%)	Dispatch (%)	Mean selected dispatch probability	Paired dispatch DEF shift x 10,000 [95% CI]
GAT, seed 42	96.6	3.4	0.0085	0.000 [0.000, 0.000]
GAT, seed 43	88.4	11.6	0.0368	-20.358 [-32.488, -8.371]
GAT, seed 44	90.4	9.6	0.0211	-0.011 [-0.029, 0.000]
GAT-Outage, seed 42	91.4	8.6	0.0228	+0.088 [+0.047, +0.131]

Checkpoint	No-op (%)	Dispatch (%)	Mean selected dispatch probability	Paired dispatch DEF shift x 10,000 [95% CI]
GAT-Outage, seed 43	94.6	5.4	0.0127	-4.403 [-8.368, -1.285]
GAT-Outage, seed 44	92.8	7.2	0.0181	-0.044 [-0.092, +0.004]

Figure 4.9 visualizes the imbalance and the paired dispatch estimates.

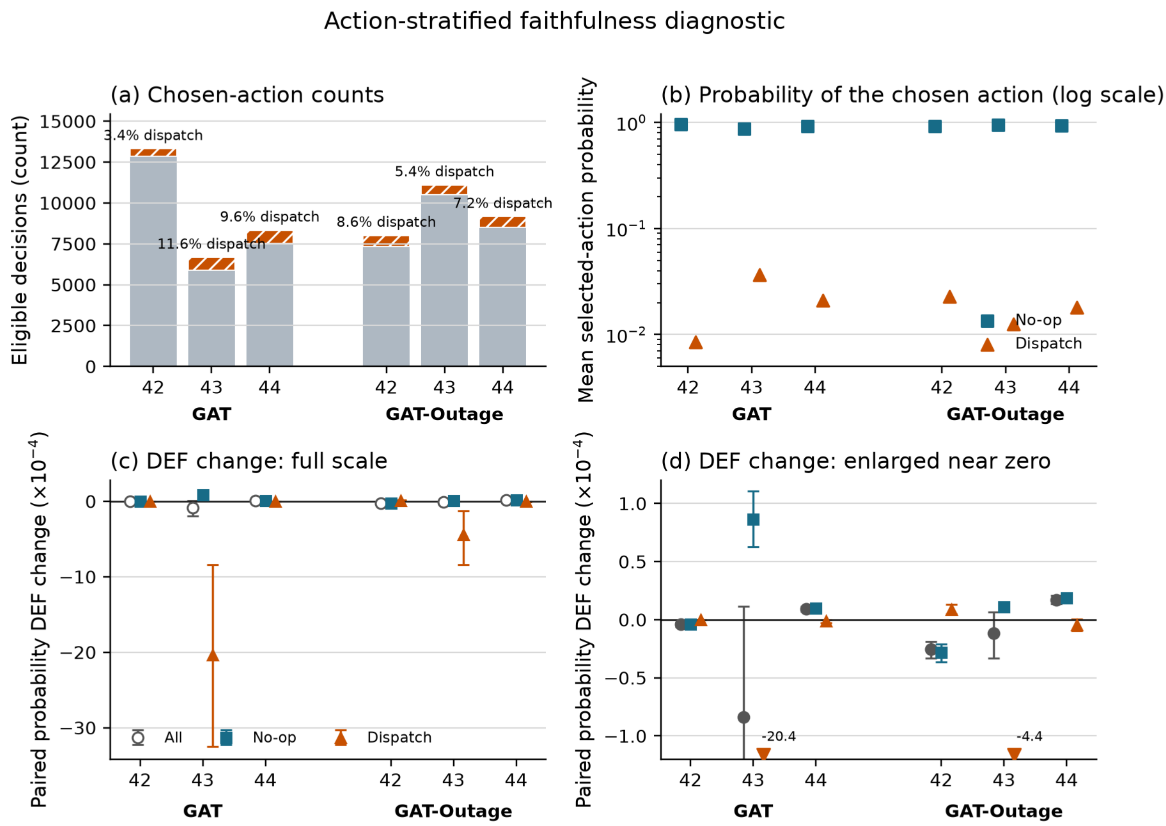

Figure 4.9. (a) Absolute no-op and dispatch counts; labels give the dispatch share. (b) Mean probability assigned to the sampled action on a logarithmic scale. (c) Paired degraded-minus-clean probability DEF on the full scale. (d) Enlarged view around zero; arrows mark intervals that extend beyond the panel. Error bars are 95% episode-block bootstrap intervals.

The no-op stratum closely follows the combined result because it accounts for most records. The same H1 trend does not appear in the dispatch stratum for any checkpoint (p = 0.44-1.00), and the dispatch results show no consistent H4 evidence. The paired dispatch DEF shift is negative in four checkpoints, positive in one, and exactly zero in one. In four of six checkpoints its direction differs from the combined estimate. These effects are small except for the seed-43 checkpoints, and dispatch intervals are wider because fewer episodes contain scored dispatch actions. The diagnostic does not establish a separate pattern for dispatch decisions. It shows that the combined H1 and H4 findings should not be generalized to dispatch actions.

4.8 Random-trigger sensitivity analysis

The random-loss control replaces the fixed tunnel-entry trigger with an independent per-taxi trigger while retaining the same 30-second observation-layer freeze. The trigger probability is fixed across checkpoints, so realized exposure remains dependent on the trajectory induced by each policy. Random-to-tunnel exposure-rate ratios range from 0.44 to 0.98, with a median of

0.78. The comparison therefore focuses on the direction of paired effects within stale-exposed decisions rather than treating the two conditions as perfectly exposure matched. Table 4.7 reports the trigger comparison.

This sensitivity analysis is a separate 24-episode evaluation. Its repeated 30-second tunnel estimates therefore differ slightly from the 48-episode main sweep in Table 4.5; no checkpoint is retrained or reselected.

Table 4.7: Tunnel-triggered and random-loss sensitivity results

Checkpoint	Exposure tunnel/random (%)	Attention shift tunnel/random (values x 1,000)	Probability DEF shift tunnel/random (values x 10,000)
GAT, seed 42	0.580 / 0.569	+4.281 / +4.037	-0.012 / +0.004
GAT, seed 43	1.176 / 0.517	+0.231 / -0.286	-0.479 / +7.361
GAT, seed 44	0.818 / 0.532	-3.623 / -3.165	+0.020 / +0.322
GAT-Outage, seed 42	0.967 / 0.713	+4.279 / +5.453	-0.205 / -0.093
GAT-Outage, seed 43	0.584 / 0.480	+0.958 / +0.444	-0.088 / -0.014
GAT-Outage, seed 44	0.712 / 0.648	-3.063 / -4.296	+0.091 / +0.518

Figure 4.10 shows the tunnel and random-trigger sensitivity results.

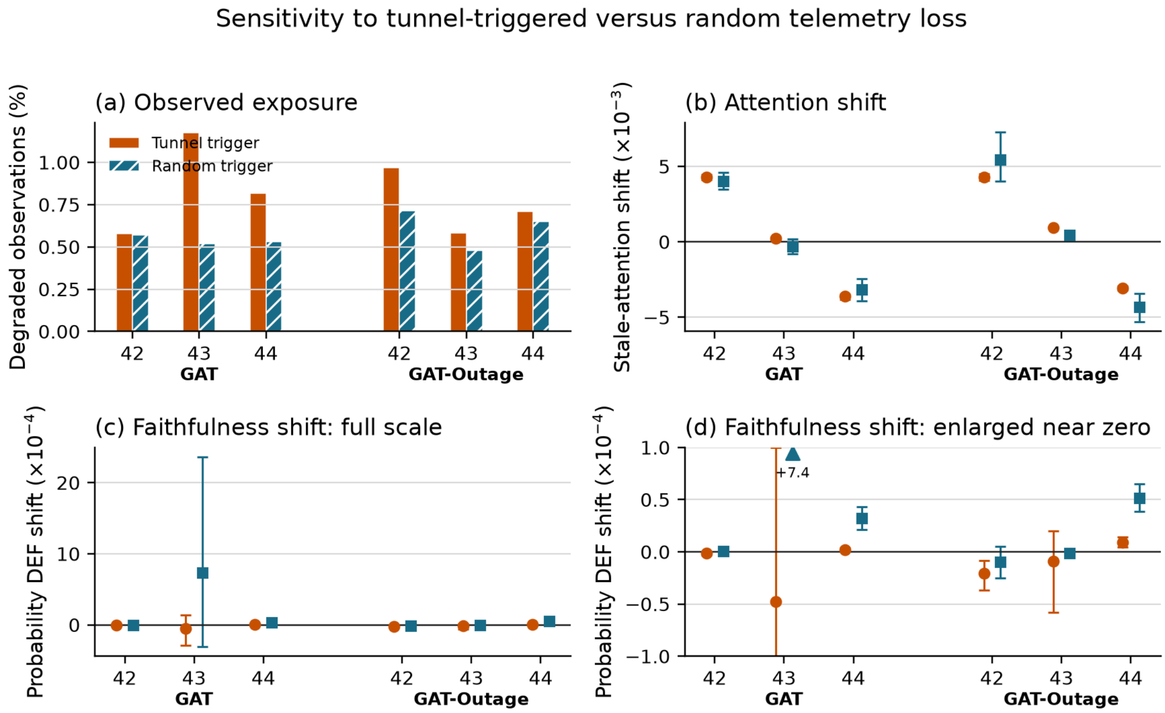

Figure 4.10. Sensitivity to tunnel and random triggers under a 30-second freeze: (a) exposure rate; (b) stale-attention shift; (c) probability-DEF shift at full scale; and (d) enlarged DEF near zero. Arrows mark off-panel intervals; error bars show 95% episode-block bootstrap intervals.

Attention-shift direction agrees between tunnel and random triggers in five of six checkpoints. Probability-DEF direction agrees in four of six. One mismatch for GAT seed 42 consists of values very close to zero. GAT seed 43 is the clear exception, but its random-trigger DEF interval is wide and crosses zero. The seed-42 positive attention pattern and seed-44 negative pattern otherwise remain visible under random triggering in both training regimes. This reduces the likelihood that the overall checkpoint dependence is produced only by the chosen tunnel location. It does not establish a location-independent effect, because the realized exposure rates and stale-node populations are not identical.

4.9 Faithfulness hypotheses

The event-aware audit uses every decision with visible stale exposure rather than a periodic sample. H1 is supported for two of three GAT checkpoints and all three GAT-Outage checkpoints. H4 is supported for all six: within-episode WAMSN-DEF correlations range from -0.199 to -0.084 for GAT and from -0.423 to -0.212 for GAT-Outage. H3 remains consistently supported. Figure 4.11 shows the per-checkpoint correlations, while Table 4.8 summarizes all hypothesis outcomes.

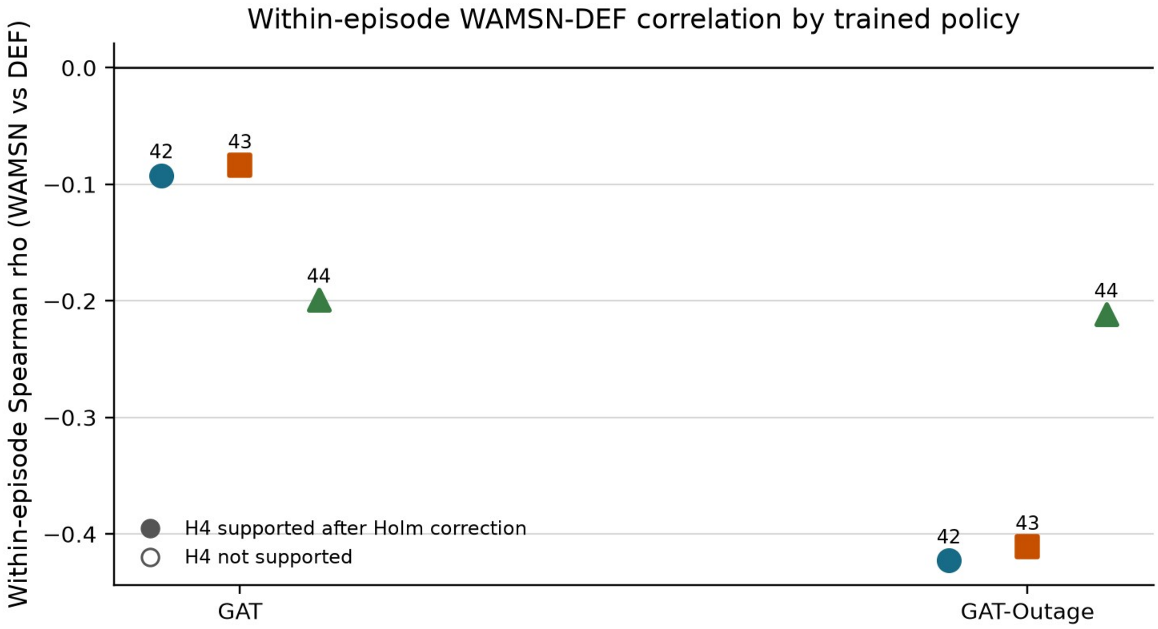

Figure 4.11. All checkpoints have a negative within-episode WAMSN-DEF association after Holm correction; the direct paired effect remains mixed.

Table 4.8: Hypothesis outcomes across training seeds

Hypothesis	GAT seeds supporting	GAT-Outage seeds supporting	Verdict
H1: outage duration increases, DEF decreases	2/3	3/3	strong but not universal
H2: faithfulness declines faster than performance	1/3	3/3	model-dependent; exploratory
H3: outage duration increases, WAMSN increases	3/3	3/3	supported as a stale-exposure manipulation check
H4: higher WAMSN is associated with lower DEF	3/3	3/3	consistently supported within episodes
H5: degradation-aware training reduces the adverse DEF relationships	n/a	0/3	not supported

For H5, every matched GAT-Outage checkpoint has more negative H1 and H4 correlations than its clean-trained GAT counterpart. The H1 comparisons for seeds 42, 43, and 44 are -0.164 versus -0.069, -0.141 versus -0.058, and -0.025 versus -0.017. The corresponding H4 comparisons are -0.423 versus -0.092, -0.411 versus -0.084, and -0.212 versus -0.199. None meets the requirement that both relationships become weaker, giving 0/3 support for H5.

H1 and H4 are association tests within one trained policy and primarily describe no-op decisions in this action-imbalanced sample. They do not by themselves establish that AoI causes lower faithfulness or that the same relationship holds for dispatch actions. The paired clean-twin audit supplies the more direct intervention contrast, and its sign still changes with training seed. H2 is treated as exploratory because clean DEF is close to zero and the relative-rate formulation is unstable.

4.10 Result summary

Longer outages increase stale exposure, while greater WAMSN accompanies lower within-episode DEF. Yet seed reversals, no-op dominance, dispatch failures, and random-trigger exceptions mean that raw attention cannot provide a reliable indication of data freshness or explanation faithfulness.

4.11 Demonstrative framework decision

The completed evidence was passed through the operational audit described in Section 3.10. Table 4.9 separates successful evidence collection from failed explanation-release checks.

Table 4.9: Freshness-aware explanation audit decisions

Audit check	GAT	GAT-Outage	Key evidence
Evidence integrity	PASS	PASS	3/3 checkpoint preflights passed for each model
Freshness reported separately	PASS	PASS	10,832 and 10,885 stale-exposed records reported
Action-aware controls	PASS	PASS	type matching, chosen-action protection, and LOO present
Dispatch decision relevance	FAIL	FAIL	clean/60 s CIs: GAT [-0.00367, -0.00252] / [-0.00371, -0.00252]; GAT-Outage [-0.00765, -0.00594] / [-0.00752, -0.00578]
No-op and dispatch composition	PASS	PASS	26,278/2,020 and 26,307/1,945 no-op/dispatch records
Attention-extraction stability	FAIL	FAIL	an alternative reverses the default direction in every seed
Checkpoint consistency	FAIL	FAIL	stale-attention direction is positive, positive, then negative across seeds
Deployment-action capability	N/A	N/A	stochastic sampling is the audited action rule; argmax is descriptive
Tunnel/random trigger robustness	INDET.	INDET.	no supported reversal, but at least one 95% interval crosses zero
Final decision	WITHHOLD	WITHHOLD	decision relevance, extraction stability, and checkpoint consistency fail

INDET. denotes INDETERMINATE: the available evidence neither passes nor fails the check.

Both models pass the checks that establish whether the evidence is complete and properly separated. Dispatch self-row margin-DEF is below the matched-random boundary under both clean and 60-second conditions, and the extraction and checkpoint checks also fail. Trigger robustness remains indeterminate because some confidence intervals cross zero. The argmax diagnostic remains an important policy limitation but is not a release gate for the sampled action rule audited here. The result is WITHHOLD for all six frozen checkpoints, so their attention maps remain internal diagnostics. This table is a demonstrative application of the post-study release rules, not a new hypothesis test or independent evidence for the main conclusion.

Chapter 5: Discussion

5.1 Answer to the central problem

The central problem is whether a complete graph-attention map can be trusted when some of its vehicle data are stale. The results show that **separate validation is required**. Raw attention alone provides no reliable evidence of freshness or decision relevance.

This conclusion is based on several results, not one p-value. Longer outages increase stale-data exposure, and the combined within-policy tests associate that exposure with lower DEF in all six checkpoints. The direct clean-twin comparison is less uniform: attention reallocation and paired DEF shift have the expected signs in their point estimates for checkpoints trained with seeds 42 and 43, but both signs reverse for checkpoints trained with seed 44 under both training regimes. The measured effects are also small.

The action-stratified results limit how broadly this finding can be interpreted. Chosen no-op actions account for 93.0% of scorable decisions with at least one available request, and the combined H1 and H4 patterns are not reproduced among dispatch actions. Four checkpoints also change paired-DEF direction between the combined and dispatch strata. The primary statistics describe the sampled action mixture, not a universal explanation response for both dispatch and no-op decisions.

The supporting controls make the interpretation stronger and narrower. The LOO-ranked control produces a positive DEF for every checkpoint, so the evaluator has some sensitivity to a perturbation-based ranking. However, expected top-k overlap reaches about 60% at k=3, which limits resolution. Taxi-only attention-LOO correlation is positive but varies substantially in strength. These checks do not show that the evaluator is perfect and attention fails. They show that, within the evaluator's measured but limited resolution, raw attention lacks reproducible evidence of explanation faithfulness. Figure 5.1 summarizes this cross-seed evidence path.

Cross-seed evidence matrix

Question	Measure	Cross-seed result	Verdict
RQ1	Clean DEF	Near type-matched random baseline	Not validated
RQ2	Paired attention shift	Positive in 4/6 trained policies	Mixed by seed
RQ3	Paired DEF shift	Negative in 4/6 trained policies	Mixed by seed
RQ4	Degradation training	No consistent mitigation	Not supported

Conclusion: attention and paired faithfulness responses are not consistent across trained policies.

Figure 5.1. Cross-seed results for the four research questions. The matrix keeps clean faithfulness, paired attention response, paired DEF response, and the effect of degradation-aware training as separate findings.

5.2 What the stale-exposure result means

WAMSN rises with outage duration for every GAT and GAT-Outage checkpoint. It is useful as an operational exposure indicator: it shows when an attention map contains more weight attached to old vehicle readings.

WAMSN is not evidence that AoI causes lower faithfulness. The metric itself weights attention by normalized AoI, so part of the increase follows directly from the manipulation. H1 and H4 show that DEF tends to be lower at greater exposure within most frozen policies. The paired audit asks a stricter question: does replacing the current observation with its degraded twin change attention and DEF at the same decision? That result changes sign across checkpoints trained with different seeds. Exposure, within-policy association, and paired intervention answer different questions and should be reported separately.

5.3 Dependence on checkpoint and analysis choice

The primary aggregation gives positive stale-attention shifts for checkpoints trained with seeds 42 and 43 and negative shifts for checkpoints trained with seed 44. The values are small: approximately -0.0035 to +0.0041 of total attention mass. Paired probability-DEF shifts are also small and reproduce the same direction in both training regimes. This shared pattern across the two model conditions suggests sensitivity to the trained checkpoint, not a stable response created by outage-aware training. The between-checkpoint range is larger than the observed difference between the two training-condition means. With only three training seeds, this is a descriptive comparison of selected checkpoints, not a variance decomposition or evidence that the seed value itself causes the response.

The random-trigger sensitivity analysis checks whether this pattern is only an artifact of the fixed tunnel location. The same attention-shift direction appears in five of six checkpoints, and the same probability-DEF direction in four of six. The seed-42 positive attention pattern and seed-44 negative pattern appear under both triggers and both training regimes. This makes a tunnel-only explanation less plausible, but it is not a complete location control. A fixed trigger probability produces different realized exposure rates after each policy changes taxi trajectories, and the identity of the stale taxis also changes. The result supports checkpoint dependence but cannot establish one general effect of random or tunnel-triggered loss.

The paired probability-DEF effects are small and may have little operational importance. The concern is not a large universal decline. Instead, even these small changes reverse direction across independently trained checkpoints, while raw corrected DEF under clean telemetry is already near zero. Together, these results do not provide a consistent basis for trusting the explanations, but they also do not show large damage from telemetry degradation.

The aggregation audit identifies a second problem. Individual heads can reverse the sign within the same checkpoint. The six checkpoint ranges extend from -0.0109 to +0.0089 across the tested reductions. Selecting another head or layer after seeing the data could change the verbal conclusion. The declared mean remains the primary result, but the sensitivity analysis shows that it is not a unique explanation produced by the network.

Query-row choice is similarly checkpoint-dependent. The selected request row improves request-action DEF by about 0.0064 for GAT seed 43 and 0.0073 for GAT-Outage seed 43, but changes little for the other four checkpoints. One query-row change therefore cannot be assumed to correct near-zero or negative DEF. An operator-facing attention map needs a declared and tested rule for choosing its query, layer, and heads.

The deterministic diagnostic reveals a related policy limitation. All six graph-attention checkpoints choose no-op in all 18 argmax episode evaluations, although their sampled policies outperform the declared lower bounds. This limitation does not invalidate the faithfulness calculations for sampled actions, but it means that the experiment audits stochastic policy decisions rather than a deployable deterministic dispatcher. The audit framework should evaluate the action rule intended for deployment and report any alternative action rule as a separate diagnostic.

5.4 Role of the construct-validity audit

The construct-validity audit supports the primary experiment. Request nodes are both information and actions; peer taxis are context only. Uniform random deletion therefore creates a large action-removal artifact. Type matching and chosen-request protection make the comparison fairer.

The LOO control checks a different question: can DEF respond when nodes are ranked by their own single-node margin loss? Its consistently positive result shows limited sensitivity, mainly for comprehensiveness. It is a positive control, not ground truth for the combined comprehensiveness-sufficiency score. Gradient x Input changes sign across checkpoints and is not automatically a stronger explanation in this task.

Together, these checks change the conclusion from a simple “attention is near random” statement to a more defensible one. Raw attention remains below the LOO control in all six checkpoints and has positive taxi-only agreement with LOO, but both DEF and rank agreement vary by trained policy. The aggregation and query-row audits add further sensitivity. This provides some evidence of decision relevance, but not enough to treat attention as reliable by default.

The top-k diagnostic further limits this interpretation. Attention-LOO overlap is above the expected matched-random overlap at k=1, but it offers no clear advantage over that baseline at k=3. This is consistent with reduced resolution as larger subsets overlap, not proof that the top attention node is the model's true reason or that every three-node display is random.

5.5 Degradation-aware training

Training with 30-second outages does not solve the explanation problem. GAT-Outage shows the same seed-dependent attention and paired DEF directions as GAT. It strengthens the H1 and H4 associations, but does not make the direct degraded-minus-clean response consistent across independently trained policies. Under the matched-seed H5 rule, both adverse correlations would need to move closer to zero. Neither does so for any of the three seeds.

These findings apply only to the tested form of robustness training. The reward contains no term for explanation stability or faithfulness, so task-reward optimization alone is not designed to align attention with a human explanation. A future intervention would need an explicit explanation objective and evaluation on independently trained models.

5.6 Proposed audit framework

5.6.1 Purpose and scope

To address this risk, the thesis proposes a **freshness-aware explanation audit framework**. The framework leaves the trained policy unchanged and does not claim to make attention faithful. Instead, it sets rules for deciding when an attention map has enough evidence to be shown as an audited candidate explanation within a stated scope. Table 5.1 links each observed risk to the required evidence and decision rule.

Table 5.1: Freshness-aware explanation audit framework

Observed risk	Required check	Evidence	Decision rule
Attention may be attached to stale data	Report freshness separately	AoI and WAMSN	Never infer freshness from attention strength
Attention may not identify decision-relevant nodes	Use action-aware, type-matched tests	DEF, margin-DEF, and LOO	Do not claim a reason without evidence of decision relevance
Combined results may be dominated by no-op	Report no-op and dispatch strata	Action counts and stratified DEF	Do not generalize a combined result across action types
The result may depend on attention extraction	Test the query row, layer, head, and aggregation	Sensitivity audits	Predeclare the extraction rule and report material sensitivity
One checkpoint may not represent another	Audit each checkpoint before release	Per-checkpoint results	Repeat after retraining or replacement
A pattern may not reproduce	Compare independent training runs	Results across seeds	Require a consistent conclusion across runs
The audited action may differ from deployment	Test the intended deployment action rule	Sampled-policy or argmax capability evidence	Require argmax capability only when deployment uses argmax
A tunnel-specific pattern may not transfer	Compare tunnel and random triggers	Paired shifts and 95% confidence intervals	Withhold when supported directions conflict or remain indeterminate

The framework is intended for offline release review before an attention map is shown to a dispatcher or other operator. It should be repeated after retraining, checkpoint replacement, changes to the telemetry or graph pipeline, transfer to a new operating scenario, and during periodic model review. It is not a live freshness monitor. A deployed system must still display freshness separately and suppress or warn about explanations based on stale inputs. Figure 5.2 shows the corresponding offline audit workflow.

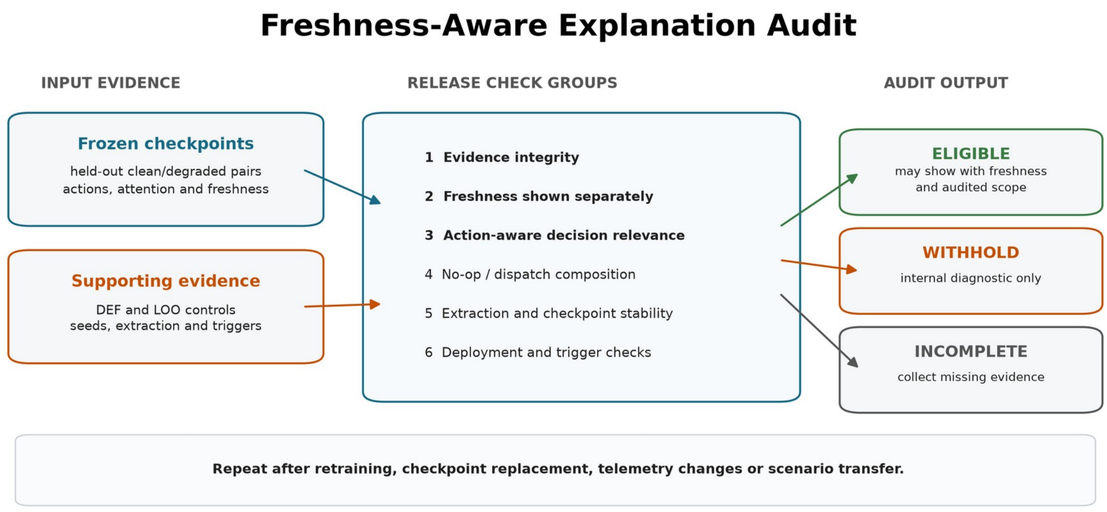

Figure 5.2. Saved evidence from frozen checkpoints passes through nine release checks, grouped here into six categories. The output controls whether attention may be presented as an explanation; it does not alter the model or selected action.

5.6.2 Inputs, outputs and use

The framework accepts evidence rather than raw attention alone. Table 5.2 summarizes its inputs and outputs.

Table 5.2: Audit framework inputs and outputs

Component	Contents	Role in the audit
Candidate scope	frozen checkpoints, model names, training seeds, and held-out conditions	defines exactly what the decision covers
Freshness evidence	paired clean/degraded records, AoI, stale-node masks, WAMSN, and stale-attention shift	separates data age from attention strength
Faithfulness evidence	type-matched and action-protected DEF, LOO, and dispatch-action results	tests whether highlighted nodes matter to the action
Stability evidence	action strata, extraction alternatives, checkpoint synthesis, deployment-action diagnostic, and trigger comparison	tests whether the conclusion survives relevant choices
Audit output	per-check status, per-checkpoint decision, model decision, failed reasons, and permitted use	supports a traceable explanation-release decision

The operating procedure is:

1. freeze the candidate checkpoint and evaluation protocol;
2. generate paired freshness, action-aware faithfulness, action-composition, and sensitivity evidence on held-out demand;
3. run preflight to establish that the evidence is complete and valid to analyze;
4. apply every release check to each checkpoint and then compare independent training runs; and
5. present attention externally only if the result is ELIGIBLE, together with its freshness indicator and audited scope.

A preflight PASS is not an explanation-release pass. It only confirms that the evidence can be analyzed. The release output is ELIGIBLE, WITHHOLD, or INCOMPLETE. Individual checks may also be INDETERMINATE when the evidence is complete but does not support a direction. A failed or indeterminate required check produces WITHHOLD; missing evidence produces INCOMPLETE. In both cases, attention must not be presented as the reason for the action.

5.6.3 Application to this study

Applying the framework to the completed evidence gives WITHHOLD for GAT and GAT-Outage and for all six individual checkpoints. This is not a failed experiment. The audit has completed and has found that the evidence is insufficient for explanation release. In particular, the dispatch-action decision-relevance intervals do not exceed the matched-random boundary, attention extraction can reverse the stale-attention direction, the stale-attention response changes direction across training seeds, and the tunnel/random comparison is indeterminate because several confidence intervals cross zero. Every argmax diagnostic also produces zero pickups, but this is a descriptive limitation rather than a release gate for the sampled action rule.

Sampled dispatch performance provides evidence of task capability, but it cannot show that the displayed reason is current or decision-relevant.

The framework checks whether an explanation has enough evidence to be shown; it does not make the model itself more faithful. Improving the model would require an explicit explanation objective and independent evaluation, as discussed in Section 5.8.

The current study demonstrates the framework's rejection path but does not show that a real trained model can reach ELIGIBLE. The LOO positive control shows that the evaluator can reward a more informative ranking, while software tests confirm that complete passing evidence produces an ELIGIBLE decision. A future faithfulness-aware model is still needed to validate the eligibility path empirically. The current thresholds must not be weakened after seeing these results simply to obtain a passing model.

5.7 Limitations

The study uses one simulated district, 20 taxis, and 50 requests. Tunnel exposure remains scenario-dependent, although the event-aware audit scores every observed stale-exposed decision. It evaluates three independent training seeds per model. Three seeds reveal variation but cannot estimate the wider distribution of training outcomes. The selected policies pass the training stability gates and their stochastic evaluations outperform random and greedy lower bounds. However, every selected GAT checkpoint collapses to no-op under the small deterministic check. The study therefore cannot establish deterministic deployment capability or isolate the contribution of graph edges to sampled capability.

The local graph is small. Type matching makes random top-k sets overlap heavily with the explanation, which compresses DEF. LOO is based on the same single-node perturbation as comprehensiveness and is not an independent ground truth explanation. Other perturbation choices or post-hoc explainers may give different results.

The degradation layer freezes position and speed for fixed windows. Real telemetry may include delayed packets, partial failures, map-matching errors, and asynchronous recovery. The study audits attention as an explanation; it does not claim that attention is useless inside the policy computation.

The paired clean/degraded comparison supports internal validity for the tested observation-layer change because both observations come from the same simulated traffic state. Simulation and scenario-design bias remain. Road-network choice, generated demand, vehicle behavior, and tunnel placement may affect which taxis become stale and how those taxis are positioned in the local graph. In particular, a fixed tunnel can create an exposure-selection effect if its traffic conditions differ from the rest of the network. The conclusions are therefore limited to the tested SUMO setting and should not be read as population estimates for real fleets or cities.

5.8 Future work

Future experiments should add training seeds and use multiple road networks, demand levels, tunnel placements, and public mobility traces. The present random-trigger control should be extended to several calibrated exposure rates and persistent loss processes so that trigger location, exposure frequency, and outage duration can be separated. Experiments should also include disconnected-edge and message-passing ablations to establish how much the policy uses its graph channel. Delayed, intermittent, and biased telemetry should be compared with fixed freezes.

The explanation audit should also compare LOO with graph-specific post-hoc methods and evaluate objectives that directly reward explanation consistency. Any proposed improvement should be tested on held-out checkpoints, not only on more episodes from one trained model. A human-factors study could then test whether explicit freshness warnings lead operators to interpret the map more appropriately.

Chapter 6: Conclusion

This dissertation examined a practical assurance problem: a graph-attention map can remain visible and appear complete after some vehicle telemetry has become stale. Attention strength alone does not tell an operator whether its source is current or whether the highlighted node affected the action.

The results lead to a clear conclusion: **this experiment does not validate raw graph-attention weights as reliable explanations under clean or degraded telemetry**. This is not a claim that every attention map is wrong. It means that attention cannot be assumed trustworthy by default.

The four research questions are answered as follows:

- **RQ1: Is attention faithful under clean telemetry?** The six checkpoints do not show a consistent result. Raw-attention DEF ranges from -0.0009 to +0.0031 across checkpoints. LOO produces a larger positive result for all six checkpoints, while taxi-only attention-LOO rank correlation ranges from +0.088 to +0.699.
- **RQ2: Does attention move toward stale vehicle nodes?** Not consistently. Under the predefined aggregation method, attention shifts toward stale nodes for checkpoints trained with seeds 42 and 43 and away from them for seed 44 under both training regimes. The same attention-shift direction appears under random triggering in five of six checkpoints, while individual heads can still reverse the direction within one checkpoint.
- **RQ3: Does the explanation remain reliable as outage duration increases?** The evidence is not consistent across analysis levels or action types. H1 is supported in five of six selected checkpoints and H4 in all six, but the combined direct paired DEF shift is negative for four checkpoints and positive for two. The effect is small and reverses for the checkpoints trained with seed 44. Moreover, 93.0% of scorable decisions with at least one available request are no-op, and the H1/H4 patterns are not reproduced in the dispatch stratum.
- **RQ4: Does degradation-aware training improve explanation reliability?** Outage training provides no consistent improvement. GAT-Outage shows the same seed-dependent paired direction as GAT.

This interpretation depends on the construct-validity audit. Deleting a request can also delete an action, while deleting a peer taxi only hides context. Type-matched and action-protected controls remove much of this artifact. The positive LOO result shows that DEF has some perturbation sensitivity, but the high expected top-k overlap limits its resolution. The conclusion draws on several checks rather than one near-zero number.

The 30-second random-trigger sensitivity analysis reduces concern that the checkpoint-dependent pattern is produced only by the selected tunnel. It does not remove the need for multiple tunnel placements or real telemetry traces, because realized exposure and the stale-node population still differ between trigger mechanisms.

The performance results also have an important limit. Sampled policies outperform the basic lower bounds, but all six graph-attention checkpoints choose no-op in 18 of 18 deterministic diagnostic episodes. The study audits explanations for sampled stochastic actions; it does not demonstrate a deployable argmax dispatcher.

In this thesis, **faithfulness decoupling** refers to the gap between a visible explanation and valid supporting evidence. An attention map can outlive the freshness of its inputs without giving a stable indication of decision relevance. The gap does not require DEF to decline monotonically with outage duration.

The freshness-aware explanation audit framework addresses this gap. It reports freshness and action composition separately, uses action-aware and type-matched tests, checks the attention extraction rule, audits every candidate checkpoint, and requires consistency across independent training runs. Stable dispatch output is not an explanation test.

Its demonstrative application gives WITHHOLD for GAT, GAT-Outage, and all six frozen checkpoints. This means that the audit completed but the attention maps did not meet the release conditions. An ELIGIBLE model is not required to support the present finding: correctly withholding unsupported explanations is the purpose of the framework. However, the study validates only the rejection path on trained models. A future model with an explicit explanation objective is needed to test the eligibility path on held-out evidence.

An ELIGIBLE result would permit attention to be shown only as an audited candidate explanation within the stated scope, with freshness displayed separately. It would not prove a complete causal explanation.

The framework is designed to reduce operational risk, not to remove faithfulness decoupling from the model. Attention remains an internal diagnostic unless the full audit provides consistent evidence.

List of Publications

No publication is claimed in this dissertation.

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Appendices

Appendix A: Experiment Reproduction

The complete reproduction procedure is provided in docs/REPRODUCE_EXPERIMENTS.md. Compact citable outputs are stored in the version-controlled results directory identified in that guide.

Appendix B: Supporting Artifacts

The repository contains the experiment configuration, source code, validation selections, preflight reports, per-seed analyses, summary tables and figure sources used in this dissertation. Raw cells and checkpoints remain outside the document because of their size.